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SPACE SYSTEMS COMMAND

SPACE SYSTEMS COMMAND
INSTRUCTION 13-610

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SPACE ACCESS

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This instruction implements DAFPD13-6, *Space Policy* and provides guidance on Roles & Responsibilities in accomplishing the Space Access Core Function. This instruction applies to all SSC military, government service, and applicable contractor personnel, as outlined within the contract, whose duties directly relate to the management, operations, preparation, and conduct of activities required to support the SSC mission. This instruction does not apply to the Air Force, Air National Guard, or Air Force Reserve. Wherever this instruction is inconsistent with current contracts that support SSC's mission, the contract shall govern. Wherever this instruction is inconsistent with DAFI 63-101/20-101, *Integrated Life Cycle Management,* the higher-level DAF guidance shall govern. Send comments and suggested improvements on a DAF Form 847, *Recommendation for Change of Product*, through appropriate command channels to SFPAE/SA S3/5, 10351 Samuel Phillips Parkway, Building 85125 CCSFS, Patrick SFB FL 32920, **ssc.aa2345.workflow@us.af.mil**. Organizations requesting document changes should ensure all units that could be affected by the change are included as informational addresses. Subordinate organizations are encouraged to supplement this instruction. Supplements will be coordinated through appropriate command channels and approved by SFPAE/SA S3/5 prior to the supplement certification and approval process. Supplements will not lessen the requirements nor change the basic content or intent of this instruction. Ensure that all records generated as a result of processes prescribed in this publication adhere to AFI 33-322, *Records Management and Information Governance Program,* and disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. The authorities to waive delta/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See DAFMAN 90-161,

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SUMMARY OF CHANGES

This document has been totally rewritten, incorporates changes in response to the establishment of the United States Space Force (USSF), Space Systems Command (SSC), Space Force Portfolio Acquisition Executive Space Access (SFPAE/SA), Space Launch Delta 30 (SLD 30), Space Launch Delta 45 (SLD 45), and Space System Delta (SYD 80) and should be reviewed in its entirety.

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Chapter 1

OPERATIONS OVERVIEW

1.1. Presentation of Forces & Capability.

1.1.1. The string of launch failures in the late 1990s highlighted a fragmentation of authority, responsibility, and accountability for safe delivery of spacecraft to orbit. The 1999 Broad Area Review (BAR), convened to evaluate these failures, strongly recommended the acquisition command be assigned and accountable for those responsibilities and authorities. That recommendation was adopted and implemented via Air Force Space Command Instruction (AFSPCI) and has been sustained, supported by multiple additional BARs, since the early 2000s. This accountability is for both spacecraft programs and launch programs and highlights a key distinction between operational chain of command and acquisition lines of authority. This document sustains this chain of accountability while adapting it from its Air Force instantiation to the current Space Force construct.

1.1.1.1. While many parallels can be drawn between current Department of the Air Force (DAF) acquisition commands, SSC (and SMC before it) has a unique role: while both USAF and USSF acquisition commands develop, acquire, and sustain air or space capabilities (respectively), SSC also has the responsibility to field those space capabilities (ref. JP 3-14, *Space Operations*). Further, with the advent of the Space Force, the two launch base organizations at Vandenberg Space Force Base (VSFB) and Patrick Space Force Base (and Cape Canaveral Space Force Station) (PSFB/CCSFS) are incorporated within SSC thus uniting the responsibility and accountability for delivering space capability to orbit under a single Field Command (FLDCOM).

1.1.1.2. Three factors were considered in designating the space acquisition command Commander as the single authority to assess readiness for launch for the Department of War (DoW) launch enterprise: (1) the inability to correct serious flaws in either launch vehicle or spacecraft after liftoff; (2) the significant operational consequences of a mission failure prior to delivery of on-orbit assets to the operating organization; and (3) the high cost of certain exquisite missions.

1.1.2. SSC executes service retained and operational warfighting space access missions. PAE/SA secures and provides reliable and responsive launch services, oversees and manages launch and test infrastructure, and delivers servicing, mobility and logistics capabilities that operate in, from and to the space domain and is responsible for pioneering rapid space access, mobility and logistics capabilities. SSC SLDs operate, maintain, and sustain systems and infrastructure to execute spaceport operations enabling the full spectrum of space launch and test activity on their installations in furtherance of national security and economic objectives.

1.2. Launch & Spaceport Overview.

1.2.1. Assured Access to Space. Title 10 United States Code (USC) § 2273, *Policy regarding assured access to space: national security payloads*, specifies the U.S. must ensure, to the maximum extent practicable, that the United States has the capabilities necessary to launch and insert United States national security payloads into space whenever such payloads are needed in space. Assured access to space and rapid deployment of space assets are foundational elements of the operational flexibility required for multi-domain dominance.

This shall include, to the maximum extent possible, resources and policy guidance to sustain the availability of at least two space launch vehicles (or families of space launch vehicles) capable of delivering into space any payload designated by the Secretary of War or the Director of National Intelligence as a national security payload; a robust space launch infrastructure and industrial base; and the availability of rapid, responsive, and reliable space launches for national security space programs to improve the responsiveness and flexibility of a national security space system; and maintain risks of mission success at acceptable levels. These key concepts support and implement National Security Strategy, National Space Policy, National Space Transportation Policy, National Military Strategy, and DAF directives. These policies indicate assured mission capability for critical space systems can be achieved only through assured and protected access to space. Assured access to space requires close partnership with commercial launch service providers and other federal agencies, such as the National Reconnaissance Office (NRO), Federal Aviation Administration's Office of Commercial Space Transportation (FAA-AST), and the National Aeronautics and Space Administration (NASA).

1.2.2. National Space Transportation Policy. National Space Transportation Policy (21 Nov 13) directs the Secretary of Defense (Secretary of War (SW) 2025), as launch agent, to maintain the capacity to evolve and support the space transportation systems, infrastructure, and support activities necessary to meet national security requirements. Department of Defense Instruction (DoDI) 3100.12, *Space Support*, allocates responsibility for National Security Space (NSS) transportation, including launch, to the Secretary of the Air Force. The USSF executes launch and spaceport operations through SSC on behalf of SecAF. SSC's spacelift mission is to ensure successful and safe delivery of space systems to support national objectives and provide a decisive advantage to US forces worldwide.

1.2.3. Spacelift. Spacelift is a key enabler that establishes and supports a broad range of space capabilities. The successful accomplishment of the spacelift mission is built on a foundation of processes and procedures refined over the years. The US uses launches to populate satellite constellations on a scheduled basis and on an emergent basis when required. Launches are planned for and executed within the Current Launch Schedule Review Board (CLSRB) process. NSS launches are predominantly used to deploy and sustain constellations, but are also used to support, deploy, or advance a variety of national security mission needs.

1.2.4. Licensed Government Launches. FAA-AST's licensing authority does not apply to launches by and for the Government (Title 51 USC § 50919, *Relationships to other Executive Agencies, Laws, and International Obligations*). In some cases, however, USG agencies may choose to contractually require FAA-AST licensing for launches of USG missions. The determining factor for licensing is not payload or mission ownership, but rather the level of engagement the government (i.e. Program Office, contracting government entity or SLD) undertakes over the course of the launch service. For this reason, licensing of launches should be reserved for missions where the USG has a higher risk tolerance, limited operational need, and/or does not intend to substantially control the launch and have limited involvement in pre-launch activities. For those launches in which the government is substantially involved as to be effectively directing or controlling the operation such that it transfers liability from the LSP to the government, the licensing approach is not a viable action. The actions of the involved government agency alone are determinant for when a launch may be licensed and therefore

any decisions to impose, or waive, a processing requirement or standard should carefully be considered before taking any action.

Chapter 2

GENERAL RESPONSIBILITIES

2.1. Spacelift Operations.

2.1.1. SSC is the lead command for DoW spacelift operations. SSC/CC delegates safety, range/spaceport operations, and base operating support elements to the SLD commanders. The SLD/CCs will take actions to eliminate duplicative requirements and approvals for FAA-AST licensed launch and reentry operations in support of the development of a robust space launch industrial base. The FAA-AST is the federal agency with regulatory authority for licensed launch activity. SLD/CCs have an inherent authority and legal responsibility for all activity that occurs on their installations. The SLDs will work in partnership with the FAA-AST to accomplish USG objectives. FAA-AST licensing processes and the DAF-FAA Launch and Reentry Activity Memorandum of Agreement (MOA), (June 2021), allow commercial launch providers to use USSF processes and services to meet licensing obligations provided they meet FAA requirements. DoW will continue to consult on the launch provider's license IAW Title 51 USC § 50918, *Consultation*, to ensure national security interests are preserved. Consultation, as acknowledged by the FAA Administrator in the DAF-FAA MOA, is an ongoing process that may continue up to T-0 and be expressed by the SLDs' representative to the FAA at any time. Additionally, the SLDs will perform ongoing day-to-day coordination at the launch site for LV/SV processing and provide launch support services IAW agreements entered for spaceport support and property usage IAW DAF standards. Consistent with the SecAF policy memo for providing launch support services to commercial entities, the SLD/CC may require specific installation services be utilized if necessary to meet safety, security, or operational efficiency needs.

2.2. Launch.

2.2.1. Launch Readiness Responsibilities: All launch agencies operating on USSF spaceports will confirm launch readiness to the SLD/CC (or designee) prior to initiating Day of Launch (DoL) activities. Those processes will be sufficiently robust to provide the SLD/CC with a level of confidence that launch vehicle performance and hazard mitigations present an acceptable risk to the public, spaceport personnel, property, operations, and security. The SLD commander will receive the launch agency's launch readiness certification no later than the Launch Readiness Review (LRR), or equivalent process.

2.2.2. Launch activity is divided into three categories: (1) USG non-FAA-AST licensed launch, (2) FAA-AST licensed launch, and (3) Test and Evaluation (T&E). Each category processes launch vehicle and space vehicle mission assurance responsibilities consistent with launch and spacecraft procurement responsibilities.

2.3. Spaceports.

2.3.1. SSC operates the Launch and Test Range System (LTRS) on USSF spaceports to meet launch requirements, ensure launch safety, and provide for safe passage of vehicles to, though, and from space.

2.3.2. USSF Spaceports support all users consistent with national policy and guidance. USSF supports commercial launch activity under guidance provided in Title 10 USC § 2276, *Commercial Space launch Cooperation*, and Title 10 USC § 2276a, *Special Authority for*

Provision of Space Launch Support Services to increase space launch capacity. The SLD/CC has the authority to prioritize activity on the installation. This includes prioritizing schedules to best meet the installation needs. Changing scheduled activity is a support interruption IAW signed agreements and is not preemption, as defined in Title 51 USC § 50910, *Preemption of Scheduled Launches or Reentries*.

2.3.3. Space activity by a commercial space entity is not commercial T&E use of the range as provided for in Title 10 USC § 4175, *Use of Test and Evaluation Installations by Commercial Entities*. A private U.S. entity engaging in space activities is considered a "commercial space entity." The commercial space entity shall hold a valid Commercial Space Operations Support Agreement (CSOSA) to be permitted to utilize USSF Real Property, resources, and support services for their space activities. Prior to signing a CSOSA, SFPAE/SA S3/5 will conduct an assessment, in coordination with the installation Plans and Programs Office, to determine if the private U.S. entity is engaged, or intends to engage in space activity. Further details are provided in **Attachment 5**, *Commercial Space Operations Support Agreements.*

2.3.4. Commercial T&E users can also request access to the range. When supporting commercial T&E, criteria of 10 USC § 4175, as implemented by DoDD 3200.11, *Major Range and Test Facility Base (MRTFB),* will be used. Companies requesting access to T&E infrastructure under 10 USC § 4175 provisions will make a request to the installation, and the installation will coordinate, if necessary, with SFPAE/SA S3/5 on the support to be provided to avoid any conflicts.

2.3.5. SLD/CCs have an inherent and delegated responsibility to ensure activities on SSC spaceports/installations are conducted safely, and an explicit responsibility for resource protection. Resource protection may be provided through a combination of insurance, indemnification and risk mitigation activities. These obligations extend to all operations on the range, services provided, and preservation of national security interests.

2.3.6. For FAA-AST licensed launches, the FAA has a statutory authority and regulates the commercial provider to ensure public health and safety, the safety of property, and national security and foreign policy interests of the United States under Title 51 USC, *NATIONAL AND COMMERCIAL SPACE PROGRAMS*. The FAA holds the FAA-AST licensed provider responsible for the safety of any licensed launch/reentry activity.

2.3.7. Services provided by SLDs in support of FAA-AST licensed activity will be performed IAW USSF processes and standards. If service is solely to meet a user requirement (e.g. not to meet a spaceport/installation safety or security requirements), it can be waived IAW user processes.

2.3.8. Use of airfields by launch and reentry vehicles (RV) is a service and shall be considered a space activity. For commercial entities, launch and reentry activity on a USSF airfield shall be covered by the CSOSA-directed relationship.

Chapter 3

ORGANIZATIONAL ROLES AND RESPONSIBILITIES

3.1. Space Systems Command (SSC) Commander.

3.1.1. The SSC Commander has overall responsibility and authority for safety, range/spaceport operations, and base operating support. SSC/CC has Space Flight Worthiness Certification (SFWC) responsibility (SFWC Authority can be delegated as necessary) for all SSC-procured non-licensed launch systems, CSO-designated Department of the Air Force (DAF) satellite systems and the integrated Launch Vehicle/Space Vehicle (LV/SV) stack.

3.1.2. Responsible for SFWC processes and procedures.

3.1.3. Chairs (or delegates) Flight Readiness Reviews (FRRs) for SSC-procured missions.

3.1.4. Delegates authority for safety, range/spaceport operations, and base operating support elements to the SLD/CC.

3.1.5. Approves (or delegates authority) Mission Director (MD) Training & Certification Plan.

3.1.6. Certifies an MD for USG launch of SSC-procured spacecraft or other DoW mission partner spacecraft on SSC-procured non-FAA-AST Licensed launch services. Assigns on-scene authority to the MD for SSC-procured spacecraft on SSC-procured, non-FAA-AST licensed launch services. The NRO will appoint, train, and certify MDs for their missions.

3.1.7. Mission Director (MD) Responsibilities:

3.1.7.1. MDs are appointed and certified by the SFWC Authority (SSC/CC or delegate) for all SSC-procured launch systems and when applicable CSO-designated DAF satellite systems and are trained in accordance with approved National Security Space Launch (NSSL) or Rocket Systems Launch Program (RSLP) MD training plan. The NRO appoints, trains, and certifies MDs for NRO missions. MDs for other Launch Agencies are appointed by those agencies. For Intercontinental Ballistic Missile (ICBM) launches, refer to AFGSCI 99-102, *Intercontinental Ballistic Missile (ICBM) Operational Test and Evaluation* (OT&E), for MD roles. The MD may select key personnel as on-console advisors.

3.1.7.2. Monitors SV and LV processing and ensures systems meet SFW criteria. Ensures all LV and SV anomalies affecting flight worthiness are successfully mitigated/resolved. Presents status to SFWC Authority at Flight Readiness Review (FRR).

3.1.7.3. MDs will participate in De-/Re-Manifest Board for assigned missions (as applicable).

3.1.7.4. Responsible for integrated LV/SV stack and carries SFWC Authority on console.

3.1.7.5. Reports mission readiness to the SLD/CC (or designee) at LRR.

3.1.7.6. Leads DoL LV & SV/RV teams through launch execution, including SV/RV separation.

3.1.7.7. Exercises waiver authority for user requirements.

3.1.7.8. Ensures a thorough Post-Flight Review is conducted for every mission to assess the performance of the LV, satellite, and ground systems.

3.1.8. Coordinates (or designee) on all staff packages for orbital debris mitigation requirement deviation and exception requests IAW **Attachment 3**.

3.1.9. Serves as USSF lead command for coordinating DAF support to commercial space activities (DAFPM2026-10-01).

3.2. Directorate of Safety (SSC/SE).

3.2.1. Adviser to SSC/CC, SLDs, and SYD 80 for launch and range safety matters affecting operations on SSC spaceports.

3.2.2. Develops standardized safety policy and guidance for SSC operated spaceport activities, including but not limited to launch/reentry operations, failure investigation and design/testing analysis.

3.2.3. Oversees and supports launch/spacecraft mishap investigations on behalf of SSC/CC.

3.2.4. Reviews/coordinates System Safety Management Plans, High and Serious Risk, and Risk Management Policies.

3.3. SSC Spacecraft Programs.

3.3.1. Assign launch-site SV program support and SV Mission Assurance (SVMA) responsibilities and payload recovery data (support Payload Recovery Team (PRT) efforts) to SLD 30/45 via Letters of Assignment (LOAs). Consults with and provides signed copies of LOAs to appropriate SYD 80 Launch Vehicle Product Line and SFPAE/SA S3/5 as applicable.

3.3.2. Develops and implements processes to ensure space vehicle(s), hosted payload(s), and related ground system(s) meet SFW requirements. Processes will be sufficiently robust to provide: a level of confidence the SV or payload presents an acceptable risk to Do-No-Harm requirements; launch environment survivability; and successfully performs over its design life.

3.3.3. Meet standard interface specifications and other applicable compatibility requirements ISO enterprise launch execution IAW **Attachment 10**. SV Interface and Orbital Debris Mitigation Standard Practices (ODMSP) standards apply to non-SSC Space Force and other DoW mission partners.

3.3.4. Responsible for SVMA and SFWC recommendation to SFWC Authority.

3.3.5. Ensures Range Safety requirements are coordinated with applicable SLD Safety Office during the lifecycle of the SV program.

3.3.6. Coordinate individual spacecraft technical and schedule requirements with SFPAE/SA S3/5 for incorporation (or inclusion) into consolidated PPF schedule. Information will be provided during recurring Data Calls or changes to ILC date impacting PPF scheduling.

3.4. Space Force Portfolio Acquisition Executive, Space Access (SFPAE/SA).

3.4.1. Secures and provides launch services, operating and sustaining launch and test infrastructure and Data Architecture, delivering space-based capabilities to, from, and through the space domain.

3.4.2. Approves New Entrant Launch Service Providers (LSP) NSSL certification plans IAW USAF Launch Services New Entrant Certification Guide and certifies LSPs, including new/modified LVs or family of LVs, are qualified to launch operational national security spacecraft.

3.4.3. Provides oversight of Space Access Manifesting Process and provides final Space Access manifest approval.

3.4.4. Approval authority, in consultation with the SV customer, for launch service provider usage for missions involving higher risk tolerance and reduced Government-performed mission assurance requirements. Includes launch service licensing (FAA-AST license) determination for USSF-funded launch activities to ensure operational equities are preserved.

3.4.5. Ensures achievement of validated capability and performance requirements for NSSL missions.

3.4.6. Develops launch and spaceport requirements through appropriate DAF requirements processes.

3.4.7. Responsible for developing, coordinating with applicable DAF/DoW agencies and advising on spaceport modernization, expansion and utilization, to include statutory, regulatory, and policy considerations.

3.4.8. In coordination with SAF/IE and SAF/GC, establishes DAF positions and policy on National Environmental Protection Act and Real Property actions such as launch site allocation, launch capacity expansion, and leases/licenses on USSF spaceports.

3.4.9. Provides comprehensive framework for spaceport strategic and long-term planning.

3.4.10. Ensures SV capability documents are consistent with validated spacelift capability documents.

3.4.11. Develops and maintains spacelift architecture products.

3.5. Space Access Operations & Requirements (SFPAE/SA S3/5).

3.5.1. Develops, implements, and reviews standardized policy and guidance to SLD's supporting USG, FAA-AST licensed launches and T&E activities and provides assistance to subordinate units with compliance issues when resolution is beyond their scope and/or resources.

3.5.2. Serves as SSC lead for DAF support to commercial space activities.

3.5.3. Develops, maintains, and executes CSOSAs (IAW **Attachment 5**), authorizing commercial entities to utilize Department of the Air Force support services and/or facilities in support of space activities.

3.5.3.1. Coordinates with installation commanders (regardless of FLDCOM/USAF MAJCOM) to develop CSOSA annex for commercial entities prior to allowing installation access.

3.5.3.2. Maintains a separate CSOSA version specifically for State-level organizations defined as commercial entities in 10 USC § 2276a. Document will contain unique State requirements not applicable to traditional commercial companies.

3.5.4. Coordinates with HQ AFMC/A1 in the management of 2M0XX personnel serving within SSC.

3.5.5. Manages NSSL and RSLP Hague Code of Conduct (HCoC) notification process IAW **Attachment 4**. HCoC pre-launch notifications are transmitted to U.S. Dept of State for SSC-procured, non-FAA-AST licensed launch activities.

3.5.6. Develops, documents and provides to SYD 80/CC and SLD/CCs an enterprise prioritization framework to inform resource and launch support service allocation.

3.5.7. Develops command policy for commercial space activities IAW existing law, policy directives and international treaties.

3.5.8. Manages mission assurance SFWC processes and procedures.

3.5.9. Develops and implements a clearinghouse process to manage USSF Payload Processing Facility (PPF) capacity utilization and scheduling.

3.5.9.1. Collects technical/schedule requirements and changes to ILC dates from Spacecraft Program Offices via recurring data calls.

3.5.9.2. Provides PPF plan to affected customers to inform planning ISO spacecraft and programs activities.

3.5.9.3. Act as sole interface between DoW agencies and National Reconnaissance Office, Office of Space Launch (NRO/OSL) for agencies requesting use of NRO/OSL launch site services.

3.5.10. Serves as space access interface to the space operational community and prioritizes SV requirements based on warfighter requirements.

3.5.11. Leads the annual SV Data Call to identify and collect unknown SV requirements. Coordinates SV's requiring PRT consideration with SYD 80 and SLDs on behalf of SSC/CC.

3.5.12. Develops payload risk tolerance assessment to inform mission design.

3.5.13. Co-Chairs [with SYD 80/AM] Procurement Manifest review board. Provides SFPAE/SA with warfighter requirement driven, programmatically executable product.

3.5.14. Develops annual Launch Service Delivery Order (LSDO) identifying launch service and Multi-Mission Manifest procurement requirements for implementation by SYD 80/AM.

3.5.15. Coordinates advocacy efforts with appropriate USSF and Higher Headquarters (HHQ) to provide necessary resources (equipment, manpower, prioritization, etc.) to meet, sustain, and evolve SSC launch and spaceport capabilities, and to provide support to other external launch customers as required by directive, regulation, and Public Law.

3.5.16. Coordinates spacelift operations activities with other stakeholders, including but not limited to other DoW agencies, Major Range and Test Facility Bases (MRTFBs), NRO, Field Commands (FIELDCOMs), Major Commands (MAJCOMs), NASA, Missile Defense Agency (MDA), Space Development Agency, FAA-AST, and spaceport users to identify/mitigate operational conflicts.

3.5.17. Supports mishap investigations IAW DAFI 91-204, *Safety Investigation and Reports*, DAFMAN 91-222, *Space Safety Investigation and Reports*, AFI 51-307, *Aerospace and Ground Accident Investigations*, and FLDCOM/CC direction.

3.5.18. Act as the Office of Primary Responsibility (OPR) or Office of Coordinating Responsibility (OCR) for range and space strategies, policies, and operational requirements.

3.5.18.1. Performs oversight into organizational constructs, program management, operations, and range posture requiring space staff support.

3.5.18.2. Provide policy, guidance, and advocacy support to the Space Access Program Offices and the SLDs through board and IPT meetings for executive inquiries and responses (e.g. CRR, legislative proposals).

3.5.19. Identify and advocate for Space Access enterprise operational needs and capability gaps, and foster partnership with external organizations.

3.5.20. Implements SSIO-directed launch vehicle/spacecraft interface standards IAW **Attachment 10.**

3.5.21. Coordinates on SYD 80 and Spacecraft Program LOAs & Memorandums of Agreements (MOA) with SLDs/SLSs to validate requirements.

3.6. System Delta, Space Access, Commander (SYD 80/CC).

3.6.1. When fulfilling NSSL Program Manager duties/role:

3.6.1.1. Responsible for development of standardized LV mission assurance (LVMA) and fleet surveillance requirements for SSC procured launch systems/services and associated training programs.

3.6.1.2. Recommends Independent Review Team (IRT) focus areas for each individual mission, informed by the risk tolerance level of the mission set, to SFWC Authority for consideration.

3.6.1.3. Responsible for LVMA and SFWC recommendation to SFWC Authority.

3.6.2. Responsible for development of standardized LVMA requirements and associated training programs for RSLP missions.

3.6.2.1. Recommends IRT focus areas for each individual mission, informed by the risk tolerance level of the mission set, to SFWC Authority for consideration.

3.6.2.2. Responsible for LVMA and SFWC recommendation to SFWC Authority.

3.6.3. Conducts Launch Service Technical Insight reviews for on-ramped NSSL Lane 1 providers to assess compliance with NSSL standards and requirements.

3.6.4. Ensures correctness and completeness of NSSL System Performance Requirements Document (SPRD, NSSL-S-001, Rev C, 14 Sep 2023), and NSSL Standard Interface Specification (SIS, NSSL-S-002, Rev D, 25 Sep 2023).

3.6.5. Maintains configuration management of NSSL command media IAW NSSL Configuration Management Plan (NSSL-P-005, 1 May 2019).

3.6.6. Implements NSSL Program Risk, Issue, and Opportunity Management Plan (NSSL-P-005, 27 Oct 2023).

3.6.7. Confirms technical rigor and consistent application of mission assurance practices IAW NSSL Mission Assurance Plan (NSSL-P-027, 8 Sep 2023).

3.6.8. Ensures application of engineering processes IAW NSSL Systems Engineering Plan (NSSL-P-029, 14 Sep 2023).

3.6.9. Ensures implementation of program protection principles IAW NSSL Program Protection Plan (LE-P-003, 18 Aug 2022) and launch service provider specific annexes.

3.6.10. Oversees execution of system safety program IAW NSSL System Safety Management Plan (LE-P-006, 16 Nov 2017) and governing policies.

3.6.11. Develops standardized training requirements ISO NSSL mission execution.

3.6.12. Verifies NSSL mission compliance with U.S. Government Orbital Debris Mitigation Standard Practices IAW **Attachment 3**.

3.6.13. Oversees program execution responsibilities, including program/contract management, LVMA, SFW process, and supporting launch operations.

3.6.14. Assigns NSSL program execution, launch site LV processing/operation and New Entrant Certification responsibilities to the SLDs via LOA.

3.6.15. Establishes and coordinates [MOU, MOA, etc.] with SLDs to define specific roles, responsibilities, and CONOPS ISO NSSL mission execution.

3.6.16. [SLD 30 Only] Assigns RSLP mission execution tasks to SLD 30 via LOA.

3.6.17. Coordinates MOAs/LOAs with SFPAE/SA S3/5 to validate requirements.

3.6.18. Ensures MD(s) are trained for launch of SSC-procured spacecraft (partner spacecraft if requested).

3.6.19. Develops and maintains standardized SFW criteria, to include launch site requirements.

3.6.20. Primary interface for interagency mission deconfliction (NASA, NRO, SDA, etc.) with contracted launch service providers during mission execution.

3.6.21. Responsible for NSSL Mishap Control Center operation and consistent standards used by SYD 80 for execution across all NSSL launch services.

3.7. Launch Vehicle Product Lines (Falcon, Vulcan, New Entrant) (SYD 80/AF/V/N).

3.7.1. Responsible and accountable to SYD 80/CC for execution of assigned NSSL program execution responsibilities, including program/contract management, LVMA, SFW process, and participation in launch operations.

3.7.2. Assesses launch system technical risk from LV design, qualification, acceptance testing, assembly, and integration through Day of Launch for NSSL IAW the mission risk posture.

3.7.3. Supports MD in ensuring anomalies affecting flight worthiness are successfully mitigated/resolved.

3.7.4. Assesses space flight worthiness of all assigned NSSL SSC procured, non-FAA-AST licensed launch systems/services, IAW SFW criteria.

3.7.5. Develops and implements supplemental Product Line LVMA processes.

3.7.6. Tracks and reports, by mission, achievement of validated capability and performance requirements.

3.8. Launch Acquisition (SYD 80/AM).

3.8.1. Co-Chairs (with SFPAE/SA S3/5**)** Procurement Manifest review board and provides operationally informed, programmatically executable product.

3.8.2. Acts as Space Access interface for launch mission planning and execution for SSC funded/procured missions.

3.8.3. Leads and is accountable for Space Access Manifesting Process; develops mission sets.

3.8.4. Provides information to spacecraft developers to enable manifesting flexibility and improved launch-on-time capabilities. Information may include but is not limited: design and production of high-fidelity mass simulators; strict compliance with the NSSL Standard Interface Specification (SIS), and integration across multiple launch vehicles/systems.

3.8.5. Responsible for SV processing contractual management, including but not limited to:

3.8.5.1. Generate task order competitions, source selection planning and execution.

3.8.5.2. Ensure execution of awarded task orders.

3.8.5.3. Serve as Contracting Officer Representative (COR).

3.8.6. Implements Launch Service Delivery Order (LSDO).

3.8.7. Tracks and reports, achievement of validated capability and performance requirements

3.8.8. Lead for all launch mission SV integration activities for NSSL and RSLP missions via assignment of a Government Mission Integration Manager (GMIM) or similar position. GMIM represents SV customers' interests from SV selection to DoL, including nominally serving as on-console lead for SV requirements and mitigation when violated.

3.8.9. Delegate RSLP decommissioned ICBM booster handling and transportation responsibilities, to 2 SLS via Memorandum of Agreement (MOA).

3.8.10. The RSLP PM is responsible for LVMA and SFWC recommendation to SFWC Authority for applicable RSLP missions.

3.8.11. Authorizes the use of locally developed procedures for RSLP LV operations IAW **Attachment 8** and maintains, for historical reference, copies of procedures for the life of the program.

3.8.12. Supports MD in ensuring integration anomalies affecting flight worthiness are successfully mitigated/resolved.

3.9. Launch & Test Range Systems (SYD 80/AR).

3.9.1. Serves as LTRS Program Management Office (PMO)

3.9.1.1. Acquisition authority for the LTRS programmatic and technical baseline, as defined by the current Capability Production Document (CPD).

3.9.1.2. Ensure technical baseline changes are compatible with higher order requirements (DAFI63-101/20-101).

3.9.1.3. Advocate common engineering processes and product solutions across Ranges for common user touchpoints, data exchanges, and customer requirements, prioritizing commonality when compatible with mission needs and cost-effective.

3.9.1.4. Assigns LTRS project execution and technical baseline management responsibilities to the SLDs via LOA.

3.9.1.5. Establishes and coordinates [MOU, MOA, etc.] with SLDs to define specific roles, responsibilities ISO Eastern and Western Range LTRS Operation and Maintenance.

3.9.2. Maintain standard Systems Engineering policies and tools (configuration management, cybersecurity, and risk management).

3.9.3. Perform program level Product Support Manager (PSM) duties and provide support to advance system health for Eastern and Western Spaceports IAW section 2.7 in DAFI 63-101/20-101.

3.9.4. Manage the overall LTRS Planning, Programming, Budgeting and Execution (PPBE) process.

3.9.5. Develop, execute, and fund LTRS PMO contracts that support system acquisition to ensure SLDs are appropriately resourced to execute responsibilities on behalf of the LTRS PMO.

3.9.6. Provide Program Management Review to PAE Space Access of expenditures, shortfalls, Unfunded Requirements, and technical decisions, as required.

3.9.7. Responsible for LTRS Cyber Strategy, Title 40 USC Subtitle III, *Information Technology Management* (Clinger Cohen) compliance, and Risk Management Framework compliance.

3.10. Servicing Mobility & Logistics (SYD 80/AL).

3.10.1. Responsible and accountable to SYD 80/CC to execute Servicing, Mobility, and Logistics activities on behalf of including but not limited to program management, contract support, service acquisition, and capability sustainment support for Point-to-Point Delivery and On-Orbit Servicing, Mobility, and Logistics systems and/or services.

3.10.2. Assess Point to Point Delivery system technical risk from Payload Delivery System design, testing, and integration, and sustainment and operations from Celestial Point of Embarkation (CPOE) to Celestial Point of Debarkation (CPOD).

3.10.3. Assess On-Orbit Servicing, Mobility, and Logistics system technical risk from a client 'do-no-harm' perspective. As a procurement of services, mission assurance equities belong to the service provider except in cases where Government Furnished Launch (GFL) is provided. In the case of GFL spacecraft design, qualification, environmental testing, and integration will be assessed from a "do-no-harm" perspective to ensure mission objectives of the combined stack.

3.11. Space Launch Delta (SLD 30/45) Responsibilities.

3.11.1. SLD/CC is accountable to SSC/CC for safety, range/spaceport operations, and base operating support. IAW **Attachment 6**. As the spaceport owner/operator and service provider,

the SLD/CC has an obligation to perform with a common law duty of care and protect the public and DAF resources.

3.11.2. Coordinate with NSSL PM to resource and execute LVMA and SFW and New Entrant Certification support IAW NSSL LOA.

3.11.3. Coordinates [MOU, MOA, etc.] with SYD80/CC to define specific roles, responsibilities, and CONOPS ISO NSSL mission execution.

3.11.4. [SLD 30 Only] Resource and execute assigned RSLP tasks IAW RSLP LOA.

3.11.5. Conducts an LRR IAW **Attachment 6** prior to any launch/landing occurring from a USSF spaceport. **(T-2)**

3.11.6. As Launch Decision Authority (LDA), provides Range clearance to launch and carries public safety authority on console. (Authority does not extend to directing an LSP to violate their FAA-AST license requirements.)

3.11.7. Supports SFWC Authority Readiness Review process IAW **Attachment 2**.

3.11.8. Execute responsibilities and provide services to commercial entities IAW **Attachment 5. (T-2)**

3.11.9. Execute real property transactions for government units and commercial entities in accordance with Air Force Civil Engineer Center (AFCEC) guidance as applicable. **(T-2)**

3.11.10. Publishes annual spaceport 5-yr strategic master plan. **(T-2)**

3.11.10.1. Outline spaceport infrastructure sustainment/modernization, real property utilization and other areas need to maximize spaceport utilization including infrastructure management.

3.11.10.2. Notify relevant government stakeholders when Real Property Agreements drive changes to tenant Blast Damage Areas and/or resource protection. **(T-2)**

3.11.11. Operates the spaceport in support of USG and commercial space launch activity IAW DoDI 3100.12 and DAFPD 13-6. **(T-0)**

3.11.12. Operates as part of the DoW MRTFB by providing government facilities and range services to all authorized government T&E users IAW DoDD 3200.11 and DoDI 3200.18, *Management and Operation of the Major Range and Test Facility Base* (MRTFB). **(T-0)**

3.11.13. Ensures spaceport readiness for launch operations and verifies operations are safe to proceed. For flyback and reentry to DAF property, (SLD/CC or designee) ensure the readiness of regardless of launch location. **(T-2)**

3.11.14. Ensures operational support is provided to authorized programs and spaceport users as documented by the Range Commanders Council (RCC) 501-12, *Universal Documentation System* (UDS) or other processes as agreed to between the SLD and spaceport user. **(T-2)**

3.11.15. Approves mission flight rules for USG and T&E operations. Reviews/approves mission rules for FAA-AST licensed operations IAW launch provider's agreement with the spaceport. **(T-2)**

3.11.16. Enforces launch commit criteria as provided in DAFI 91-202, *The Department of the Air Force (DAF) Mishap Prevention Program* and SSCMAN 91-710V1-7, *Range Safety User Requirements Manual*. **(T-2)**

3.11.17. Ensures the protection of personnel, resources, and space environment by closing the range for offending activity that violates risk criteria during launch/reentry operations.

3.11.18. Develops local spaceport support contracts with support from Spaceport Launch Squadron, contracting, and other functions. **(T-2)**

3.11.19. Ensures launch telemetry frequencies, bandwidth and power as well as trajectory data information, which includes specific primary and backup launch windows provided by the launch provider, is provided to Air Force Spectrum Management Office (AFSMO) and SSC/BZZS 60 business days in advance of planned launch date for all missions, modifications to submitted data will be sent as soon as possible. 60-day requirement is notional depending on mission type/standard trajectories. **(T-2)**

3.11.20. Ensure Critical Asset Protection assessments, notifications, mitigations, and schedule deconflictions are conducted IAW **Attachment 9** for launches/reentries conducted from their respective ranges. **(T-2)**

3.11.21. Ensures early participation of Explosive Ordnance Disposal (EOD), Emergency Management, and Fire Emergency Services in the launch planning process. **(T-2)**

3.11.22. Provide Payload Recovery Team support in the event of a launch related mishap and planned exercises as directed. **(T-2)**

3.11.23. Develop plans and identify resources, to include assigning Payload Recovery Team Director to perform space vehicle component recovery and support investigation activities, for NSSL/RSLP missions identified by FLDCOM/CC. **(T-2)**

3.11.24. Executes LTRS projects assigned by the LTRS PMO. **(T-2)**

3.12. Spaceport Launch Squadron (1/4 SLS) Responsibilities.

3.12.1. Operate and maintain the Eastern and Western Ranges to include deconfliction of installation operations, logistics, and transportation events. **(T-3)**

3.12.2. Operate Spaceport Command and Control Center (SCCC).

3.12.3. Executes LTRS projects assigned by LTRS PMO. **(T-2)**

3.12.4. Ensure Technical Directors exercise technical management authority of the physical systems and coordinate technical baseline changes with the LTRS PMO to ensure parity between the physical and programmatic baseline.

3.12.5. Supports the LTRS Program Manager in the management of LTRS technical baseline as defined by the current Capability Production Documents (CPD) and SFPAE/SA Portfolio Vision and Requirements, Volume 1: Eastern and Western Ranges, ensuring technical baseline changes are compatible with higher order requirements.

3.12.6. Define capability requirements and establish priorities IAW of DAFI 63-101/20-101.

3.12.7. Appoint a local Chief of Maintenance (focus on sustainment) and Lead Engineer (focus on architecture and modifications) to execute duties as defined in DAFI 63-101/20-101 in collaboration with LTRS PMO. **(T-2)**

3.12.8. Perform IT Service Management (ITSM) implementation, operational use, and day-to-day execution in collaboration with the LTRS PMO. **(T-2)**

3.12.9. Provide SLD-specific LTRS strategic plans, roadmaps, Program Objective Memorandum (POM) inputs, and annual spend plans to the LTRS PMO for PPBE process.

3.12.10. Perform project engineering management, execution, installation, integration, testing and transition for projects assigned through the LTRS PMO spend plan process.

3.12.11. Provide Contracting Officer Representative (COR) support of LTRS systems for Logistics, Configuration Management, and Cybersecurity contract surveillance.

3.12.12. Manage a Deficiency Review Board (DRB), assigning DRB co-chairs between SLDs and LTRS PMO. **(T-2)**

3.13. Space Launch Squadron (2/5 SLS) Responsibilities.

3.13.1. Execute day-to-day Mission Assurance (MA) responsibilities, utilizing Engineers and 2M0XX Mission Assurance Technicians, under the direction of NSSL PM consistent with the LOA **(T-2)**

3.13.1.1. Provide certification and processing oversight as a key launch-site component of the LV and SV SFWC process.

3.13.1.2. Ensure MA is performed by personnel who are trained, qualified, and certified, unless under the direct supervision of a trainer or certifier. **(T-2)**

3.13.2. Coordinates with SYD 80 LV Product Line Teams to assess LSP technical risk and adequacy of launch site procedures. **(T-2)**

3.13.3. Coordinates with SYD 80 LV Product Line Teams to monitor, assess, and report LSP adherence with technical procedures, general aerospace maintenance practices, and proper use of special tools and GSE during LV/SV processing, facility and infrastructure activities. **(T-2)**

3.13.4. Supports safety, security, environmental compliance, and resource protection (to include critical asset protection process requirements), for LV/SV processing and launch site equipment & infrastructure.

3.13.5. Integrates LV, SV, facility, and infrastructure operations on the installation ISO assigned spacelift missions. **(T-2)**

3.13.6. [2 SLS Only] Coordinates with SYD 80/AM the use of locally developed RSLP procedures IAW **Attachment 8**. **(T-2)**

3.13.7. [2 SLS Only] Performs maintenance, handling, transportation, and emplacement, in support of small lift launch operations as required by RSLP MOA. **(T-2)**

3.13.8. Participates in LRRs for non-FAA-AST licensed NSSL launch missions and reports LV, SV, and facilities readiness. See **Attachment 6** for information on LRR and discussion items. **(T-2)**

3.13.9. Responsible and accountable for SV mission assurance responsibilities IAW SV Program Office LOAs.

3.13.10. Develop and execute a structured training and qualification program for personnel performing LV/SV Mission Assurance duties IAW NSSL Mission Assurance Plan and applicable guidance. **(T-2)**

3.14. Range Operations Squadron (ROPS) Responsibilities.

3.14.1. Overall lead for spaceport/range support to the launch campaign process including coordinating pre-launch and launch events, and post-launch review briefs. **(T-2)**

3.14.2. Conducts spaceport operations and MRTFB T&E activities IAW **Attachment 6**.

3.14.3. Supports LTRS program office management activities. **(T-2)**

3.14.4. Supports the development of spaceport support contracts in coordination with contracting and other support functions. **(T-3)**

3.14.5. Ensures a structured training and qualification program is developed, documented, and maintained for key ROPS personnel supporting DoL operations. **(T-3)**

3.14.6. Provides operational support to authorized programs and spaceport users as defined through the UDS process or other instructions, plans, procedures, and processes as agreed to between the SLD and spaceport user. **(T-3)**

3.14.7. Serves as the SLD single point of contact for range scheduling to include asset deconfliction. **(T-3)**

3.14.8. Manages spaceport land, air, rail, and sea clearance issues for USG and T&E launches and as requested for FAA-AST licensed launch activity. **(T-3)**

3.14.9. Ensures frequency mitigation procedures are properly coordinated and scheduled, as required, IAW **Attachment 7**. **(T-2)**

3.14.10. Ensures space launch missions' GPS protection requirements are coordinated and deconflicted with DoW GPS Testing, Training or Exercise (TT&E) events and/or initiate CEASE BUZZER IAW process at **Attachment 7. (T-2)**

3.14.11. Provide functional oversight and direction to SLD Military Radar Unit (MRU). **(T-3)**

3.14.12. Provides a Program Support Manager (PSM) for users as the primary POC for spaceport support, including radar, telemetry, optics, command, communication, data processing/transfer/products and other support functions. **(T-3)**

3.14.13. Provides range engineering support for prelaunch operations (as required), T&E, and countdown. Determine instrumentation status, commit instrumentation for launch support, and direct resolution of instrumentation anomalies to ensure range systems are available to meet requirements in each launch Operations Directive. **(T-32)**

3.14.14. Advises SLD/CC on acceptance of new or modified range systems for operational use and approves shutdown of superseded operational range systems. **(T-3)**

3.14.15. Coordinates all contractor procedures that impact spaceport operations. **(T-3)**

3.14.16. [2 ROPS Only] Supports safety, security, environmental compliance, and resource protection (to include day-of-launch critical asset protection process requirements), for LV/SV processing and launch site equipment & infrastructure. **(T-2)**

3.14.17. [2 ROPS Only] Integrates LV, SV, facility, and infrastructure operations ISO assigned spacelift missions. **(T-2)**

3.15. SLD Safety Responsibilities (SLD/SE).

3.15.1. Serves as the senior safety advisor. **(T-3).**

3.15.2. Conduct Critical Asset Protection calculations, assessment, and notifications IAW **Attachment 9. (T-2)**

3.15.2.1. Coordinate with Range Users and ROPS to address impacts and possible mitigations, to include schedule deconfliction.

3.15.2.2. Verify day-of-launch compliance with critical asset protection risk criteria.

3.15.3. Provide recommendation to waiver authority for waiver packages including safety requirements.

3.15.4. Perform public safety risk analysis, to include DoL, IAW applicable support agreements. **(T-3)**

3.15.5. Ensures hazardous operations, range safety instrumentation, and flight safety systems meet range safety requirements IAW appropriate guidance.

3.15.6. Ensures Interim Safety Board (ISB) President is trained in appropriate tasks, roles and responsibilities of an ISB IAW DAFI 91-204. **(T-3)**

3.15.7. Provides personnel for contingency response support/Emergency Operations Center (EOC) duties. **(T-3)**

3.16. Weather Responsibilities (30 OSS & 45 WS).

3.16.1. Conducts weather support for spaceport operations IAW the supported agencies operational directive, DAF instructions/manuals, and/or the SLD weather support plan. **(T-3)**

3.16.2. Provides weather services as required for launch and reentry services. **(T-3)**

3.16.3. Ensures a robust, structured training and qualification program is developed, documented and maintained for weather personnel supporting spaceport operations. **(T-3)**

3.16.4. Provides input into the UDS process. **(T-3)**

PHILIP A. GARRANT
Lieutenant General, USSF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

10 USC § 2273, *Policy regarding assured access to space: national security payloads*

10 USC § 2276, *Commercial Space Launch Cooperation*

10 USC § 2276a, *Special Authority for Provision of Space Launch Support Services to Increase Space Launch Capacity*

10 USC § 2667, *Leases: Non-excess Property of Military Departments and Defense Agencies*

10 USC § 4175, *Use of Test and Evaluation Installations by Commercial Entities*

14 CFR Chapter III*, Commercial Space Transportation*

40 USC Subtitle III, *Information Technology Management*

51 USC § 50504, *Use of Government Facilities*

51 USC § 50902, *Definitions*

51 USC § 50910*, Preemption of scheduled launches or reentries*

51 USC § 50918, *Consultation*

51 USC § 50919, *Relationship to other executive agencies, laws, and international obligations*

51 USC*, NATIONAL AND COMMERCIAL SPACE PROGRAMS*

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JP 3-14, *Space Operations*, 6 Jan 2009

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National Space Transportation Policy, 21 Nov 2013

NSSL (EELV) System Safety Management Plan, LE-P-006, 16 Nov 2017

NSSL Configuration Management Plan, NSSL-P-005, 1 May 2019

NSSL Mission Assurance Plan, NSSL-P-027, 8 Sep 2023

NSSL Program Protection Plan, LE-P-003, 18 Aug 2022

NSSL Program Risk, Issue, and Opportunity Management Plan, NSSL-P-005, 27 Oct 2023

NSSL Systems Engineering Plan, NSSL-P-029, 14 Sep 2023

RCC 501-12, *Universal Documentation System*, Jul 2012

SMC-G-1202, *Space Flight Worthiness Certification Criteria*, 7 Oct 2009

Space Access Portfolio Vision and Requirements, Volume 1: Eastern and western Ranges

SSCMAN 91-710V1-V7*, Range Safety User Requirements Manual*, 27 Dec 2022

Standard Interface Specification (SIS), NSSL-S-002, Rev D, 25 Sep 2023

System Performance Requirements Document (SPRD), NSSL-S-001, Rev C, 14 Sep 2023

United States Government Orbital Debris Mitigation Standard Practices, 10 Nov 2019

USAF Launch Services New Entrant Certification Guide, 27 Oct 2011

Prescribed Forms

None

Adopted Forms

DAF Form 847, *Recommendation for Change of Product*

Abbreviations and Acronyms

2M0XX—Air Force Specialty Code Prefix – Missile and Space Systems

AFC—Area Frequency Coordinator

AFSMO—Air Force Spectrum Management Office

AFSS—Autonomous Flight Safety System

CC—Commander

CCSFS—Cape Canaveral Space Force Station

CFC—Space Force Field Command – Combat Forces Command

CFR—Code of Federal Regulations

CLSRB—Current Launch Schedule Review Board

CSOSA—Commercial Space Operations Support Agreement

CSpOC—Combined Space Operations Center

CD-O—Operations, Deputy Commander

CD-S—Installation Support, Deputy Commander

DAF—Department of the Air Force

DAFI—Department of the Air Force Instruction

DAFMAN—Department of the Air Force Manual

DAFPD—Department of the Air Force Policy Directive

DIRLAUTH—Direct Liaison Authorized

DNRO—Director, NRO

DoW—Department of War

DoDD—Department of Defense (War 2025) Directive

DoDI—Department of Defense (War 2025) Instruction

DoL—Day of Launch

EOC—Emergency Operations Center

EOD—Explosive Ordnance Disposal

EoLP—End of Life Plan

ES—Eastern Spaceport

EtP—Exception to Policy

FAA—Federal Aviation Administration

FAA-AST—Federal Aviation Administration Office of Commercial Space Transportation

FLDCOM—Field Command

FOCI—Foreign Ownership, Control or Influence

FRR—Flight Readiness Review

FTS—Flight Termination System

GPS—Global Positioning System

GMIM—Government Mission Integration Manager

HCoC—Hague Code of Conduct

HHQ—Higher Headquarters

HQ—Headquarters

IAW—In Accordance With

IRT—Independent Review Team

ISB—Interim Safety Board

ISO—In Support Of

KSC—Kennedy Space Center

LDA—Launch Decision Authority

LLCC—Lightning Launch Commit Criteria

LOA—Letter of Assignment

LRR—Launch Readiness Review

LSDO—Launch Service Delivery Order

LSP—Launch Service Provider

LTRS—Launch and Test Range System

LV—Launch Vehicle

LVMA—Launch Vehicle Mission Assurance

LWO—Launch Weather Officer

MAJCOM—Major Command

MD—Mission Director

MDA—Missile Defense Agency

MOA—Memorandum of Agreement

MOU—Memorandum of Understanding

MRTFB—Major Range and Test Facility Base

MRU—Military Radar Unit

NASA—National Aeronautics and Space Administration

NLT—No Later Than

NoD—Notice of Deviation

NRO—National Reconnaissance Office

NRO/OSL—NRO Office of Space Launch

NSSL—National Security Space Launch

ODMSP—Orbital Debris Mitigation Standard Practices

OASW—Office Assistant Secretary of War

PPF—Payload Processing Facility

POV—Probability of Violation

PRT—Payload Recovery Team

PSG—Program Support Guide

PSM—Program Support Manager

RCO—Range Control Officer

RE—Range Engineer

ROC—Range Operations Commander

ROPS—Range Operations Squadron

RSLP—Rocket Systems Launch Program

RV—Reentry Vehicle

SCN—Satellite Control Network

SDAR/EoLP—Space Debris Assessment Report/End of Life Plan

SE—Safety

SECW—Secretary of War

SFW—Space Flight Worthiness

SFWC—Space Flight Worthiness Certification

SFRC—Space Force Range Contract

SIB—Safety Investigation Board

SLD—Space Launch Delta

SLS—Space Launch Squadron

S/MFCO—Senior/Mission Flight Control Officer

SMO—Spectrum Management Office

SSC—Space Force Field Command - Space Systems Command

SV—Space Vehicle

SVMA—Space Vehicle Mission Assurance

SW—Secretary of War

SWS—Space Warning Squadron

T&E—Test and Evaluation

UDS—Universal Documentation System

US—United States

USAF—United States Air Force

USC—United States Code

USG—United States Government

USSF—United States Space Force

VSFB—Vandenberg Space Force Base

WS—Western Spaceport

Office Symbols

SSC/BZES—Spectrum & Space Environment Branch

SSC/CC—Commander, Space Systems Command

SSIO—SSC Organization, Space Systems Integration Office

SYD 80—System Delta, Assured Access to Space

SYD 80/AF—Falcon System & Operations Division

SYD 80/AL—Space Mobility & Logistics

SYD 80/AM—Mission Solution System Program

SYD 80/AN—New Entrant System & Operations Division

SYD 80/AR—Range Systems Division

SYD 80/AV—Vulcan System & Operations Division

SYD 80/CC—Commander, System Delta, Assured Access to Space

Terms

Aircraft—A vehicle whose use is intended for operation within the airspace and regulated under the Federal Aviation Act. A launch or reentry vehicle, as defined under the Commercial Space Act, during all phases of flight to include when transiting airspace is not considered an aircraft.

Autonomous Flight Safety System (AFSS)—An independent self-contained flight system onboard a launch or reentry vehicle with configurable rule-based software logic which replaces the traditional human decision-making process for flight safety.

Base Operating Support—Base operating support includes traditional installation support elements such as security, fire protection, disaster control, etc., required to provide, protect, and maintain facilities and supporting infrastructure.

Boilerplate—Applicable to CSOSA/Annex – Standardized language applicable to commercial space activities support.

Certification Maintenance—Surveillance of certified LSP activities to verify the risk baseline has not significantly changed from the baseline established at the time of certification.

Commercial Entity or Commercial—As defined in Title 10 USC § 2276a the term 'commercial entity' or 'commercial' means non-Federal entity organized under the laws of the United States or of any jurisdiction within the United States.

FAA-AST Licensed Launch—Launch or reentry activity licensed by the FAA. This includes activity required to prepare, conduct, or recover from a launch or reentry action when requesting services, commodities, or resources from DAF under the provisions of Title 10 USC § 2276a.

Critical Asset—From DAFI 91-202, "A resource requiring protection. It normally includes property/infrastructure that is essential to protect the public health and safety, maintain the minimum operations of the range, or protect the national security or foreign policy interests of the United States (see DAFI 91-102)."

Critical Payload—A Payload and essential infrastructure actively supporting such a payload that is a critical asset that is so costly or unique that it cannot be readily replaced or the time frame for its replacement would adversely affect the national interests of the United States.

Current Launch Schedule Review Board (CLSRB)—SFPAE/SA S3/5 chaired forum attended by space enterprise stakeholders to review resource, satellite, and launch assessments to resolve schedule/resource conflicts. Address launch base conflicts/challenges affecting government manifest execution (SV processing, multi-agency linkages, range/spaceport considerations, industry partner limitations/impacts, policy). Launch planning review to identify potential conflicts ahead of execution year (strategic initiatives, strategic planning, launch queue adjustments).

Eastern Spaceport (ES)—The ES is part of the national launch range facilities, operated by SLD 45 and located at PSFB, Florida; the spaceport includes the operational launch and base support facilities located at CCSFS, Florida; owned or leased facilities on downrange sites such as Ascension; as well as the Jonathan Dickson Missile Tracking Annex (Jupiter, Florida), and in the context of launch operations, the Atlantic Ocean. For reentries/flyback at the ES, the ES may be expanded to involve the land, sea and air space within the reach of the RV during its descent until it impacts or is recovered. Launch vehicle and payload prelaunch and launch activities are concentrated at CCSFS, Kennedy Space Center (KSC), and outlying support locations. Launch activities conducted by ES personnel operating outside the geographical limits described above may occur under DoW or USSF direction or under the auspices of agreements made by these agencies. In such cases, the term ES is expanded to include prelaunch, recovery, launch area, recovery area and impact area.

Fleet Surveillance—Surveillance of commercial and civil activities for the purpose of maintaining awareness of issues that intersect or may intersect LV flight worthiness or other government mission assurance requirements for certified Launch Service Providers and New Entrant Launch Service Providers.

Flight Readiness—Assessment of the ability of the entire launch system [LV/SV, critical facilities/infrastructure, range systems, and Aerospace Ground Equipment (AGE)] to meet the current launch schedule.

Flight Readiness Review (FRR)—Provides SFWC Authority with hardware and software mission status for the LV, the satellite, and/or critical ground systems, as well as associated interfaces, including FWC and ODMSP compliance or approved EtP by the OASW/SECW. Required for all missions where the SSFWC Authority is responsible for the certification of the mission, LV, satellite or critical ground system, and FRR is presented to the SSC/CC or a designated representative. The briefing takes place following SV and LV integration.

Flight Worthiness—Measures the degree to which a spacecraft, LV, or critical ground system, as constituted, has the capability to perform its mission with the confidence that significant risks are known and deemed acceptable.

Flyback—Return of a booster during launch phase for recovery and potential reuse. For risk and approval, the segment of flight associated with launch, will be considered a contiguous element of the launch activity. Flyback is different from reentry, which requires return from orbit.

Integration—Engineering analyses, studies and associated testing and operations necessary to support mission objectives.

Launch Activity—Activity necessary to process, prepare, place, or attempt to place, a launch vehicle and its payload, if any, in a suborbital trajectory, in Earth orbit in outer space, or otherwise in outer space. This includes static fire, booster assembly, pre-launch readiness assessments, dress rehearsals, etc.

Launch Agency—USG entity sponsoring a launch activity. The USG organization is responsible for necessary approvals and requests.

Launch Decision Authority/Landing Decision Authority—SLD/CC or delegate, provides clear to launch/land, and carries public safety authority on console. (Authority does not extend to waiving FAA-AST license requirements.)

Launch Mishap—A mishap occurring during launch vehicle operations, including upper stages. This includes payloads that do not obtain the intended orbit, re-contact of the payload with the upperstage/launch vehicle, or collisions before completion of the initial drift orbit, and flight safety systems failures.

Launch Service Provider—The commercial entity providing space launch capability and responsible for requesting a license and spaceport support or services. For a FAA-AST licensed launch activity the FAA holds the launch provider accountable for services and actions performed IAW the licensing process.

Launch Readiness Review (LRR)—Assessment of both the ability to meet the mission design requirements and the current launch schedule based on the sum of flight worthiness and flight readiness.

Launch Safety—Requirements, processes and procedures designed to protect the public, launch base personnel (government and contractor), and range infrastructure from the hazards associated with launch operations.

Launch Support Services—As described in 10 U.S.C. § 2276a, The term "launch support services" means the supplies, equipment, commodities, services, and construction supporting space launch, to include reentry.

Launch Vehicle Mission Assurance (LVMA)—LVMA is a technical and management process rigorously, continuously, and iteratively employed over the life-cycle of a launch system (system design through operations) to maximize mission success. LVMA encompasses system engineering, risk management, quality assurance, and program management by an experienced, stable launch agency team. LVMA is achieved through integrated developmental processes and/or independent technical assessment and requires expenditures commensurate with the criticality of the mission and the consequences of failure. LVMA includes new entrant certification, certification maintenance, and fleet surveillance.

Launch and Test Range System (LTRS)—The current program name for the combined system capabilities of USSF's Eastern Spaceport, headquartered at Patrick SFB in Florida and USSF's Western Spaceport, headquartered at Vandenberg SFB in California. The system provides the following distinct capabilities to support spacelift and T&E missions: Planning and Scheduling, Range Safety, Vehicle Tracking and Assessment, Positive Control, Communications, Weather, Area Surveillance and Timing and Sequencing.

Mission Assurance (MA)—An integrated engineering-level assessment of analysis, production, verification, validation, operation, maintenance, and problem resolution processes performed over the life cycle of a program by which an operator/user determines that there is an acceptable level of risk to employment of a system or end item to deliver an intended capability in an intended environment. The objective of the assurance process is to identify and mitigate design, production, test and operational deficiencies that could impact mission success.

Major Range and Test Facility Base (MRTFB)—The designated core set of DoW Test and Evaluation (T&E) infrastructure and associated workforce that must be preserved as a national asset to provide DoW T&E capability to support the DoW acquisition system (see DoDD 3200.11).

National Security Space (NSS)—The space-related systems, services, capabilities, and associated information networks of the DoW and the national intelligence community, or other space-related systems that support U.S. national security and enable defense and intelligence operations during times of peace, crisis, or conflict.

National Security Space Launch (NSSL)—Launch systems, services, and capabilities of the DoW that support US national security and enable defense and intelligence operations during times of peace, crisis, or conflict.

New Entrant Certification—Systems engineering process used by SSC to certify the capability of LSPs and launch systems to fulfill the requirements of the NSSL program.

Range—The term "range," when used in a geographic sense, means a designated land or water area that is set aside, managed, and used for range activities of the Department of Defense. Such term includes the following: (A) Firing lines and positions, maneuver areas, firing lanes, test pads, detonation pads, impact areas, electronic scoring sites, buffer zones with restricted access, and exclusionary areas. (B) Airspace areas designated for military use in accordance with regulations and procedures prescribed by the Administrator of the Federal Aviation Administration (10 U.S.C. §101(e)(1)). A "range" includes all associated range assets and services required for the operation of the range.

Range/Spaceport Operations—Any procedure that requires the use of range resources (instrumentation, collision avoidance processes, weather systems, etc.) The execution of operations focused on efficient and coordinated employment of all range assets and processes to enable the safe and timely launch and/or landing of payloads and test vehicles.

Range/Spaceport User—The responsible and accountable organization for range activity. In the case of USG launch activity it is the launch agency, for FAA-AST licensed launch activity it is the commercial launch provider, and for T&E activity it is the test agency.

Real Property—Land, buildings, structures, utilities, improvements and appurtenances thereto. Includes equipment attached to, and made part of, buildings and structures but not movable equipment. Primarily consisting of facilities and other non-equipment support system infrastructure.

Reentry Vehicle—A vehicle designed to return from Earth orbit or outer space to Earth, or a reusable launch vehicle designed to return from Earth orbit or outer space to Earth, substantially intact. (Title 51 USC § 50902, *Definitions,* (19))

Resource Protection—The protection of flight hardware, facilities, support equipment, or other property from damage due to mishaps.

SSC Procured Launch Service—Space Launch capability provided by a contractor to place a satellite into a specified orbit. The contractor retains ownership of all flight and ground hardware, engineering analyses, processes, and readiness decisions except as provided under contract.

Space Activity—Activity located at launch sites or USSF spaceports that are required to support space launch/reentry or designated to develop, demonstrate, or assess the capability of the space launch/reentry vehicle, payload, or associated support infrastructure/equipment. This also includes launch/reentry vehicle assembly, launch/reentry vehicle operations and control, communications, flight safety functions, and payload operations, control, and processing as well as any associated services, commodities, equipment, or infrastructure required to support this activity.

Space Flight Worthiness Certification (SFWC) Authority—The SSC/CC or delegate responsible for assessing and accepting risk and certifying integrated LV/SV stack readiness at the FRR.

Space Flight Worthiness Certification—Certification made by SFWC Authority for all SSC-procured launch systems, CSO-designated DAF satellite systems, and integrated Launch Vehicle/Space Vehicle stacks prior to launch. to ensure the spaceport user has confidence that launch vehicle, spacecraft, and ground system risks have been resolved, or are known and deemed acceptable. Launch vehicle mission assurance (LVMA) and space vehicle mission assurance (SVMA) and system safety are used to support flight worthiness certification briefing to SSC/CC at the FRR.

Space Vehicle Mission Assurance (SVMA)—SVMA is a technical and management process rigorously, continuously, and iteratively employed over the space vehicle life cycle, i.e. all activities associated with the development, testing and operation of space vehicles in orbit or deep space, to include spacecraft design, orbital operations, reentry, recovery and disposal elements. SVMA encompasses system engineering, risk management, quality assurance, and program management by an experienced, stable space vehicle agency team. SVMA is achieved through

integrated developmental processes and/or independent technical assessment and requires expenditures commensurate with the criticality of the mission and consequences of failure.

Spacelift—The ability to deliver satellites, payloads, and material into space. Spacelift includes space launch/reentry operations and range/spaceport operations.

Spaceport—A Spaceport is the geographic area of land, water, and associated airspace designated for use for launch and reentry of launch and space vehicles, to include shared supporting infrastructure and services such as launch or reentry support facilities, control centers, safety, security, emergency response, instrumentation, commodities, and associated equipment. A spaceport hosts one or more launch or reentry sites and processing facilities. Spaceports may support either vertical or horizontal launches and are classified as either federal, commercial, or private. Federal spaceports are owned by the federal government and do not require FAA licensing. Commercial spaceports hold an FAA-AST spaceport operator's license and can service multiple launch customers. A private spaceport (launch site) operates under an FAA-AST launch vehicle license and is used exclusively by a single operator.

Test Agency—The accountable agency conducting T&E activity on the range.

Test and Evaluation (T&E) Activity—T&E activity conducted by DoW in support of DoW weapon systems. Non-DoW organizations conducting test will be evaluated on a case-by-case basis. Examples of this activity are ballistic missile tests, Missile Defense Agency operational test, and aeronautical test activity. This does not include space activity.

T&E Infrastructure—The facilities, ranges and all other physical assets such as buildings, instrumentation, networks, range space, and frequency spectrum, used when conducting DoW Weapon System T&E.

Non-Licensed United States Government (USG) Launch—Launch activity by and for the government to include USSF, National Reconnaissance Office (NRO), Civil, and other DoW missions that do not occur under an FAA-AST license due to substantial government involvement so as to be effectively directing or controlling the launch activities (DOJ Decision on Licensing Under the Commercial Space Launch Act, 15 Nov 1990).

Universal Documentation System (UDS)—The Range Commanders Council prescribed documentation system. The UDS provides a common language and format for stating program, mission, and test requirements and supporting documentation.

Western Spaceport (WS)—The WS is part of the national launch range facilities, operated by SLD 30 at VSFB, California. and associated support sites and extends along the West Coast of the continental US westward through the Pacific and Indian Oceans. For reentries/flybacks intended at the WS, the WS may be expanded to involve the land, sea and air space within the reach of the RV during its descent from orbit until it impacts or is recovered. Launch and reentry activities conducted by WS personnel operating outside the geographical limits described above may occur under DoW or USSF direction or under the auspices of agreements made by those agencies. In such cases, the term WS is expanded to include these situations and apply, as required, to the specific mission, launch, recovery, launch area, recovery area and impact area.

Attachment 2

SPACE FLIGHT WORTHINESS CERTIFICATION PROCESS

A2.1. SFWC Process.

A2.1.1. Space flight worthiness certification is the process by which a prescribed set of technical practices proven to result in mission success are compared against the technical activities performed during a DAF-procured, space system's lifecycle. This minimum set of technical practices provide confidence in mission success in the following cases: the launch service will provide the SV(s) safe transportation to the SV(s)'s intended orbit, the SV(s) and hosted payload(s) will do-no-harm to LV performance and perform their mission over the intended life of the vehicle(s), and related SV ground systems perform the mission over the intended life of the vehicles. The comparison provides information on the activities that were/were not performed, and documents references to artifacts for the activities or rationale for why an activity was not performed. The comparison is accomplished by the combined DAF, SSC, and Aerospace Corporation team and provides the SFWC Authority an informed basis to certify space flight worthiness prior to launch. In addition to their support to the total process, Aerospace also provides an independent risk assessment to the SFWC Authority. The SFWC process comprises three aspects: SFWC criteria and review, Independent Technical Reviews, and the Space Launch Readiness Review Process.

A2.2. SFWC Criteria.

A2.2.1. SFWC is a technical activity assessing SV and LV readiness irrespective of programmatic cost and schedule considerations. This activity spans the design, build, integrate and test, LV/SV integration, launch, and readiness phases for on-orbit SV mission performance. MA processes involve the application of proven technical practices, leveraging risk management techniques and independent assessments to give confidence in achieving mission success. Assessments of LV and SV readiness and confidence in achieving mission success feed the SFWC criteria identified in *SMC-G-1202, Space Flight Worthiness Certification Criteria* and *SSC-G-012 Systems Safety Program Guidebook* organized into Operational Safety, Operational Suitability, Operational Effectiveness, and Mission Certification categories.

A2.2.2. Independent Reviews. A key aspect of MA and SFWC process is the use of independent reviews to provide certification authority alternative perspectives of a program under review without the same constraints as the program office. The SSC/CC Independent Review Team (IRT) carries out independent space flight worthiness assessments at SSC/CC direction, informed by risk tolerance and other applicable considerations, for risk identification and mitigation well before launch operations using a tailored, tactical, risk adjusted focus.

A2.2.3. SFWC process will leverage existing MA forums or mechanisms, to identify focus areas, which are presented to the SFWC Authority. The SFWC Authority approves the IRT mission-specific focus areas and tasks the IRT to assess these priorities, allowing ample time for responsible agency to implement any recommendations to reduce mission risk. IRT approach focuses on the following key objectives:

A2.2.3.1. Provide flexibility to focus on key MA areas based on the individual situation of each launch

A2.2.3.2. Be selectively applied based on mission risk tolerance.

A2.2.3.3. Be selectively manned with leader and technical expertise based on the focus area to be assessed.

A2.2.3.4. Focus on activities directed by the SFWC Authority with community input.

A2.2.3.5. Out-brief the SFWC Authority with a flight readiness assessment and any elevated risks identified with sufficient time to address findings/recommendations prior to, but no later than the FRR, in support of the earliest available mission Initial Launch Capability (ILC).

A2.2.4. SYD 80 SFWC Readiness Review Process. The SFWC readiness review process is a set of gated reviews where selected SFWC criteria information is presented to leadership to proceed with launch processing. Management and execution of this process is assigned to the respective mission Government Mission Integration Manager (GMIM) and includes recommendations from the LSP and program office, SV contractor and program office, appropriate SLD, IRT, and Aerospace Corporation independent assessment.

A2.2.4.1. Mission Readiness Review (MRR) - The MRR occurs at the point in time beyond which the space vehicle and launch service provider's activities cannot proceed without a significant impact to the other should any delays occur. The MRR is normally chaired by the space vehicle program manager's leadership summarizing selected content from the SFW criteria at the executive level. This review should be more detailed for the SV and related ground system readiness since it normally occurs prior to more detailed launch service readiness.

A2.2.4.2. Flight Readiness Review (FRR) - The FRR occurs just prior to launch where the SFWC Authority certifies the mission. The FRR summarizes selected content from the SFW criteria at the executive level. The FRR content is normally a subset of the information presented at the MRR with more detail on the readiness of the launch service and emphasis on the integrated stack.

A2.2.4.3. Incremental Readiness Review (IRR) - An IRR occurs when more detailed or comprehensive information needs to be presented to leadership which cannot be covered in the normal allotted readiness review meeting time, or when significant issues must be briefed well in advance of MRR or FRR to give the SFWC Authority the appropriate discretion from a risk acceptance perspective. Information that would be presented at the MRR or FRR can be briefed to leadership at an IRR. Therefore, the necessary decision-quality information is presented cumulatively to inform the decision maker.

A2.2.4.4. Post-Flight Review (PFR) – The PFR normally occurs approximately 60 days after launch. The PFR is provided to the SYD 80 representative. The PFR content should summarize launch service and space vehicle performance, describe any off-nominal behavior, and capture lessons learned for future missions.

A2.2.4.5. SFW Certification Review - In the cases where SSC/CC SFWC occurs earlier in the lifecycle than immediately prior to launch, the SFW certification review takes the place of the Flight Readiness Review. For this certification, the SFWC is a "type" certification where the identified launch service and space vehicle(s) are proven compatible. This "type" certification will occur when the SSC certification official has

confidence the space vehicle and launch service providers have a proven, reliable, process-controlled system.

A2.2.4.6. Additional detail on the above processes are provided in local SSC guidance. Contact SFPAE/SA S3/5 Chief of Space Flight Worthiness for assistance.

A2.3. Launch Activity Assurance.

A2.3.1. USG non-FAA-AST licensed launch activity can be USSF, NRO, other DoW, or Civil missions by and for the government. Activity Assurance responsibilities are shown in **Table A2.1**.

A2.3.1.1. SSC/CC is the SFWC authority for SSC-procured non-FAA-AST licensed launches. Additionally, SSC/CC will certify flight worthiness of SSC-procured, non-FAA-AST licensed launch systems/services ISO non-SSC Space Force other DoW mission partners.

A2.3.1.2. For NRO missions, Director of the NRO (DNRO) (or designated representative) will certify flight worthiness for NRO-procured spacecraft on SSC-procured launch systems/services. The SSC/CC will certify launch vehicle (LV) flight worthiness to DNRO as part of the DNRO's mission certification.

A2.3.1.3. For other DoW and Civil missions, the launch agency will perform flight worthiness certification processes. Those processes will be sufficiently robust and acceptable to provide the SLD/CC a sufficient level of confidence that launch vehicle performance presents an acceptable risk to spaceport people, property, operations, and security.

A2.3.2. FAA-AST licensed launches are launches licensed by the FAA. The FAA-AST licensing process will be used to meet SSC safety obligations for FAA-licensed launches. The FAA-AST licensing process from a federal spaceport, allows the use of safety processes as described in 14 CFR § 450.101, and the DAF/FAA Memorandum of Agreement (MOA) for Launch and Reentry. The launch providers' processes will be sufficiently robust to provide the SLD/CC a level of confidence that launch vehicle performance presents an acceptable risk to spaceport people, property, operations, security, and national security interests.

A2.3.3. T&E activity. Test activities conducted by DoW and Non-DoW entities on a MRTFB. Spacelift operations are not T&E activities. The sponsoring agency will perform launch readiness certification processes.

A2.3.4. SSC-procured hosted payloads on other organizations' SVs/LVs. The SSC certifying official will certify the SSC-procured hosted payload will do no harm to the host vehicle and meet the space vehicle interface control document (ICD) requirements. The SYD 80/AM will certify processes are robust to provide a level of confidence payload performance presents an acceptable risk to mission success over life.

Table A2.1. Launch Activity Assurance Responsibility for USSF Spaceport Operations.

		Category 1			Category 2	Category 3
		USG Non-Licensed	NRO	Civil and Other DoD	FAA-AST Licensed	T&E
ASSURANCE TASK	**SFWC (Integrated LV/SV Stack)**	SSC/CC	DNRO	Launch Agency	Launch Service Provider	Sponsor Agency
	LVMA[4]	SSC/CC	SSC/CC	SSC/CC	Launch Service Provider	Sponsor Agency
	SVMA	SSC/CC	DNRO	Launch Agency	Launch Agency[3]	Sponsor Agency
	Launch Safety[1] **Public**	SLD/CC	SLD/CC	SLD/CC	Launch Provider IAW FAA Licensing Process[2]	SLD/CC
	Launch Safety[1] **Spaceport safety protection**	SLD/CC	SLD/CC	SLD/CC	SLD/CC	SLD/CC
	Range/Spaceport Operations	SLD/CC	SLD/CC	SLD/CC	SLD/CC	SLD/CC
	Base Operations Support	SLD/CC	SLD/CC	SLD/CC	SLD/CC	SLD/CC

1 - For flyback to DAF property, the SLD/CC has a responsibility to ensure the readiness of the installation to receive the flyback portion of launch and retains the authority to provide a no-go, regardless of launch location.

2 - SLD/CC responsibility is met through FAA-AST licensing process, range user compliance with land use and other support agreements, and day-to-day interactions IAW DAF-FAA-MOA

3 - For NSS payloads launched on an FAA-AST licensed launch the Mission Director will provide the SV go/no-go status to the Launch Service Provider.

4 - SSC/CC is responsible for Space Flight Worthiness Certification for all SSC-procured non-licensed launch systems.

Attachment 3

ORBITAL DEBRIS MITIGATION

A3.1. Exception to National Space Policy. National Space Policy requires U.S. government agencies limit the creation of new debris, consistent with mission requirements and cost effectiveness, during the procurement and operation of spacecraft, launch services, and conduct of tests and experiments in space by following the United States Government Orbital Debris Mitigation Standard Practices (ODMSP). Furthermore, it assigns approval authority for exceptions to the ODMSP to the head of the sponsoring agency. For the Department of War, this is the Secretary of War. As such, the following guidance is provided to clarify roles, responsibilities and procedures for SSC offices in the event that their programs are unable to follow the ODMSP. National Space Policy applies to all organizations within the USSF, such as SSC or Space Development Agency (SDA), who are the party responsible for the LV Space Debris Assessment Report (SDAR) and SV SDAR/End of life Plan (EOLP).

A3.1.1. In the event that an SSC organization concludes their program cannot follow the ODMSP, the Deputy Assistant Secretary of War for Space Policy will be notified. SYD 80 is only responsible for the launch vehicle, not any ODMSP issues that may exist with the spacecraft. More specifically, consistent with requirements in DAFI 91-202, SYD 80 will not launch any mission partner's spacecraft on an SSC-procured launch service unless that spacecraft has been confirmed to SYD 80 as either following the ODMSP or an exception has been granted from the proper authority within the SV's program office or sponsor. This applies to any government mission partner SYD 80 is providing a launch service for, e.g., SSC, Space Development Agency (SDA), National Reconnaissance Office (NRO), National Aeronautics and Space Administration (NASA), Air Force Research Lab (AFRL), etc.

A3.1.2. When an SSC program office initially discovers they cannot follow an ODMSP requirement, or the potential risk exists that they might not be able to, the cognizant LV or SV program will promptly initiate the staffing of a notification and Exception to Policy (EtP) request through their respective Program Acquisition Executive and the Headquarters SSC staff. Prompt staffing of the request is important due to the lengthy timelines to coordinate and get approval from the Secretary of War, or to allow decision-makers in the coordination chain sufficient time to influence change that will allow compliance while not negatively affecting the launch schedule and manifest. If during the staffing process a change is made that allows the ODMSP to be followed, the EtP request can be rescinded.

A3.1.3. The request will provide a concise summary of the specific ODMSP objective and standard practice that cannot or is at risk of not being met, why following these guidelines would pose a significant risk to mission success or significant impact on cost, schedule, or performance, and a completed safety risk assessment and risk acceptance decision following standard system safety methodologies. The System Delta will coordinate the request with the SSC Space Debris Management Office (SDMO) in SSC/BZZS, the SSC Chief Systems Engineer, the SSC Director of Safety, and the SSC Commander (or designee), prior to submitting to USSF/S3I, who will then assume responsibility for staffing the request to the appropriate offices at HQ USAF, HQ USSF, SAF, OSW and finally, DASD(SP). Due to uncertainties in the amount of time required for coordination outside of SSC, program offices should develop the EtP request staff package as early as feasible.

Attachment 4

HAGUE CODE OF CONDUCT (HCOC) PROCESS

A4.1. Description.

A4.1.1. Hague Code of Conduct (HCoC) is an international agreement regarding Ballistic Missile Proliferation in which the U.S. provides pre-launch notification for all space, ICBM and SLBM launches via Department of State National and Nuclear Risk Reduction Center (NNRRC). Notification email will be marked **CONTROLLED UNCLASSIFIED INFORMATION (CUI)**.

A4.1.2. The FAA Office of Commercial Space Transportation (AST) handles notifications for FAA licensed launches, NASA handles notifications for all NASA civil launches and, the U.S. Navy handles notifications for all submarine launched ballistic missile launches.

A4.2. Pre-Launch Notification Process.

A4.2.1. SFPAE/SA S3/5 provides HCoC information to HQ Combat Forces Command (CFC Arms Control (S10T) for all NSSL non-FAA-AST licensed launches to include orbital launches utilizing decommissioned ICBM motors (RSLP).

A4.2.2. Information will be provided utilizing HQ CFC/S10T developed template (NLT 10 business days prior to launch) and must contain, at a minimum, the following.

A4.2.2.1. SFPAE/SA S3/5 Generated Control Number comprised of CY and number of launch from specific spaceport e.g. 2023001.

A4.2.2.2. Date of launch and specific vehicle being launched e.g. Falcon 9.

A4.2.2.3. Generic Class e.g. Space launch.

A4.2.2.4. Launch Area containing land/air and specific spaceport info.

A4.2.2.5. 7- day Notification period (approved launch date +6). If launch slips outside this window a new notification must be processed and labeled as RESCHEDULED.

A4.2.2.6. HQ CFC/S10T will process and provide to HQ CFC Command Center who in-turn sends to NNRRC at **NNRRC-HCOC_PLN@state.gov**; **ACN-WMC-HCOC-DL@state.gov** NLT 4-days prior to launch.

A4.2.2.7. Planned Direction (FOR LAUNCH OF A SPACE VEHICLE) and single or multiple launches.

A4.3. Responsive Launch Missions.

A4.3.1. The agency responsible for issuing the Notice to Launch order (or SFPAE/SA S3/5) will send the HCoC notification directly to NNRRC. SFPAE/SA S3/5 HCoC OPR will provide the responsible agency with a formatted notification email. Agency is responsible for providing 7-day notification period.

Table A4.1. SAMPLE EMAIL.

From: As Required
To: ACN-WMC-HCOC-DL@state.gov; NNRRC-HCOC_PLN@state.gov - Others as required
Cc: As required
Subject: CUI//Hague Code of Conduct Pre-Launch Notification for (date, vehicle) Launch from (location)
CUI
Hague Code of Conduct Pre-Launch Notification: CCSFS2023004
GENERAL CLASS: Space Launch Vehicle
LAUNCH AREA: Cape Canaveral Space Force Station, Florida
NOTIFICATION PERIOD 9 Sep 2023/0000Z to 15 Sep 2023/2359Z
PLANNED DIRECTION: FOR LAUNCH OF A SPACE VEHICLE
SINGLE OR MULTIPLE LAUNCHES: Single Launch
ADDITIONAL REMARKS: FOR INTERNAL USE ONLY
A. TRANSMIT THIS NOTIFICATION NO LESS THAN TWENTY-FOUR HOURS IN ADVANCE OF THE PLANNED LAUNCH NOTIFICATION PERIOD
B. POC FOR THIS NOTIFICATION IS (Submitter), AND THE ALTERNATE POC IS (XXXX) SHOULD THE NNRRC NEED TO CONTACT SOMEONE
C. REQUEST COMEBACK COPY OF THE NOTIFICATION AFTER TRANSMITTAL
D. REQUEST ACKNOWLEDGEMENT OF THIS EMAIL AND ANY SUBSEQUENT EMAILS BY REPLYING ALL

Attachment 5

COMMERCIAL SPACE OPERATIONS SUPPORT AGREEMENTS

A5.1. CSOSA Applicability.

A5.1.1. A commercial entity shall hold a valid USSF CSOSA and applicable SLD CSOSA Annex to be allowed access to the respective USSF installation and resources IAW U.S. Code and implementing regulations. Specifically, Title 10 USC § 2276a for launch support services and Title 10 USC § 2667, *Leases: Non-excess Property of Military Departments and Defense Agencies,* with Title 51 USC § 50504 for space related facilities. Space activity by a commercial entity is not commercial use of the Major Range and Test Facility Base (MRTFB) Test and Evaluation (T&E) infrastructure as provided for in Title 10 USC § 4175. A U.S. private entity requesting access to MRTFB T&E infrastructure under Title 10 USC § 4175 provisions will make a support request directly to the SLD and the SLD will coordinate, if necessary, with SFPAE/SA S3/5 on the support to be provided to avoid any conflicts. The spaceport and MRTFB are co-equal responsibilities of the SLD with each being derived from different authorities, directives, and policy even if some of the installation resources are used to support both operational areas. The CSOSA and associated annexes are not support agreements as defined in DoDI 4000.19, *Support Agreements* and AFI 25-201, *Intra-Service, Intra-Agency, and Inter-Agency Support Agreements Procedures,* nor do they contain provisions subject to the FAR.

A5.2. SFPAE/SA S3/5 CSOSA Process.

A5.2.1. Prior to signing a CSOSA, SFPAE/SA S3/5 will conduct an assessment, in coordination with the SLD 30/S5 and/or SLD 45/XP office, to determine if the private U.S. entity is engaged in space activity. Space activity includes activity located at launch sites or USSF spaceports that are required to support space launch/reentry or designated to develop, demonstrate, or assess the capability of the space launch/reentry vehicle, payload, or associated support infrastructure/equipment. This also includes launch or reentry vehicle assembly, launch or reentry vehicle operations and control, communications, flight safety functions, and payload operations, control, and processing as well as any associated services, commodities, equipment, or infrastructure required to support this activity.

A5.2.2. SFPAE/SA S3/5 approval of a CSOSA certifies that all activities on the installation by the commercial entity should be treated as space activity and support is in the national interest, as long as those space activities are within the scope of SLD commander approved program activities. If SLD representatives are aware of activity that they believe is inconsistent with this guidance they can remove government support from the activity of concern and query SFPAE/SA S3/5 for additional direction. Commercial entities must also have a valid CSOSA annex, and any additional documentation as required by the installation. CSOSAs, and each annex, should also be considered for use on USAF installations under the coordinating authority for commercial launch support provided by SecAF policy guidance delegating Title 10 USC § 2276a implementation to SSC. The core CSOSA and each associated annex signify an agreement that serves as the "contract or other transaction authority" of Title 10 USC § 2276a. Each installation will have an annex with the commercial entity and will contain installation determined requirements to include procedures for receiving launch support services and the funding arrangements associated with providing these services.

A5.2.3. A CSOSA is an SFPAE/SA S3/5 generated document. The CSOSA is a "boilerplate" document executed by all commercial entities requesting support from DAF installations. Minor administrative changes are permitted; however, substantive changes require approval by SFPAE/SA S3/5 and SSC/JA.

A5.2.3.1. SFPAE/SA S3/5 will maintain two versions of the CSOSA template. One "boilerplate" version for commercial entities who operate in the commercial marketplace. The other version will serve non-federal entities (10 USC § 2276a defines both categories as 'commercial entities') receiving SLD support while fostering economic development and will allow for the incorporation of unique State requirements not applicable to traditional commercial companies.

A5.2.3.2. SFPAE/SA S3/5 will maintain a list of active CSOSAs containing, at a minimum, issue/expiration date, entity active location/program and entity POC contact info.

A5.2.4. SFPAE/SA S3/5 will not execute a CSOSA (Initial or renewal) without coordination with impacted DAF Installation Plans and Programs office.

A5.2.5. Prior to CSOSA execution commercial entities operating in the commercial marketplace will be required to execute an ownership/investment questionnaire. The execution process will not proceed without executed questionnaire. The ownership/investment questionnaire will accompany request for company background vetting via government agencies and/or other contracted assessment providers.

A5.2.5.1. CSOSA applicant vetting is not limited to but will include Foreign Ownership Control or Influence (FOCI) of company management/positions of leadership and significant investors. CSOSA holders may be subject to continuous FOCI monitoring and additional vetting actions.

A5.2.5.2. If potential negative info is uncovered during vetting/continuous monitoring, SFPAE/SA S3/5 personnel will coordinate with appropriate federal agencies to assess risk, identify concerns, and provide go/no-go or rescind/suspension recommendation.

A5.2.5.3. If the recommendation/decision is to deny an initial CSOSA the opportunity for commercial entity to respond/provide corrective actions to address identified risks/concerns will be situational dependent and require SFPAE/SA S3/5 concurrence.

A5.2.5.3.1. If the recommendation/decision is to rescind, suspend or deny renewal of an existing CSOSA, SFPAE/SA S3/5 will:

A5.2.5.3.2. Provide, in writing, proposed termination/suspension detailing circumstances and timeline for commercial entity to provide, in writing, information in opposition to proposed action.

A5.2.5.3.3. If commercial entity provides information in opposition to proposed action, consult with internal/external agencies in determining course of action. Determination will be provided to commercial entity in writing. Disputes will be processed IAW appropriate CSOSA Article.

A5.2.6. CSOSA and corresponding analysis will be coordinated with SSC/JA and other federal agencies (if required) prior to sending to SFPAE/SA S3/5 for execution.

A5.2.7. Fully executed CSOSAs will be provided to the commercial entity and the supporting installation Plans and Programs office representative. CSOSAs are valid for five years from SFPAE/SA S3/5 signature date.

A5.3. Space Launch Delta (SLD).

A5.3.1. SLD's are the primary commercial space support provider. The following will apply to any DAF Installation providing commercial space support utilizing Title 51 USC or Title 10 USC authorities.

A5.3.2. SLD's will support commercial space activities IAW law, regulations, governing agreements, and policies to the maximum extent reasonable and in the best interest of the government.

A5.3.3. SLD 45/XP and SLD 30/S5 office will develop and maintain a CSOSA annex, in accordance with the CSOSA conditions, for each commercial entity the SLD/CC has agreed to support. Annex content/requirements must be provided to/acknowledged by the commercial entity prior to referral to SFPAE/SA S3/5 for an initial CSOSA.

A5.3.3.1. Annexes can be boilerplate documents preapproved by SLD agencies/leadership and applicable to all commercial entities operating on the installation. Boilerplate annexes are sufficient to meet the intent of CSOSA conditions while adequately protecting government interests and being responsive to the needs and schedule of the commercial entities. A boilerplate annex can make efficient use of existing SLD resources when coupled with SLD level guides, handbooks, instructions, and supplements. Commercial entities' acknowledgement of being presented with an annex establishes the effective date of the annex. This must occur prior to the SLD providing launch support services or facilities.

A5.3.3.2. Annexes will be developed in coordination with the other SLD to ensure as much commonality as reasonably possible while taking into consideration any SLD peculiarities.

A5.3.3.3. SLD personnel may be tasked to assist other DAF installations in CSOSA annex development.

A5.3.3.4. Annexes may be augmented by additional installation specific documents like a PSG. If so, the document should be referenced in the annex with a sufficient description for the commercial entity to understand the function and format of the additional documentation.

A5.3.3.5. SLD 30/S5 and SLD 45/XP office administers the CSOSA annex process to include maintaining list of active CSOSAs annexes containing, at a minimum, issue date, commercial entity, program name, program siting location(s) and commercial entity point of contact information.

A5.3.4. The SLD 30/S5 and SLD 45/XP will develop a local supplement, instruction, or other guidance that addresses CSOSA and Annex administration detailing specific SLD responsibilities in implementing CSOSA and Annex administration procedures. The SLD supplement, instructions or guidance will be coordinated with the other SLD and SFPAE/SA S3/5 to ensure a consistent approach to the maximum extent reasonably possible taking into consideration any SLD peculiarities.

A5.3.5. Upon request from a commercial entity for space-related facilities, property, resources, services, and/or commodities the SLD shall consider their ability to support and if such support is in the national interest. The SLD/CC may mandate the use of government provided services by the commercial entity, if the commander determines it is in the best interest of the government and contributes to the operational efficiency, safety, or security of the installation.

A5.3.6. The SLD/CC is authorized to defer, delay, modify, or otherwise interrupt support as necessary to meet priority requirements consistent with Current Launch Schedule Review Board (CLSRB) firing order. Any such resource reallocation decision is a launch support interruption as described in the CSOSA. Commanders should make changes to approved support only when necessary and make all effort to provide stable and predictable support for all users.

A5.3.7. The SLD/CC reserves the right to prohibit, suspend or terminate performance under the annex, if doing so is in the best interest of the Government.

Attachment 6

SPACEPORT OPERATIONS

A6.1. Launch Readiness Review Assessment.

A6.1.1. An LRR will be conducted prior to any launch occurring from a USSF controlled range. **(T-2).** The SLD/CC will determine the content of the LRR. The following are recommended attendees:

A6.1.1.1. SLD/CC (or designated representative chairs the LRR)

A6.1.1.2. MD

A6.1.1.3. LV System Program or AM, as applicable

A6.1.1.4. SV program office for appropriate missions

A6.1.1.5. SLD/CD-O

A6.1.1.6. SLD/SE

A6.1.1.7. SLD/CD-S

A6.1.1.8. FAA Representative for FAA-AST licensed launches

A6.1.1.9. Contractor Representatives (e.g. Launch service provider, Range Operations & Maintenance, SV provider, etc.)

A6.1.1.10. Civil Agency Representative for Civil missions.

A6.1.2. The following items are recommended for LRR content:

A6.1.2.1. Launch Personnel. Review personnel status of key launch positions to include the certification status.

A6.1.2.2. Mission Description.

A6.1.2.3. Schedule. Identify all major milestones of all missions leading up to launch and affecting the scheduled launch.

A6.1.2.4. Launch Facility/Platform. Review the status and any remaining open items or major discrepancies.

A6.1.2.5. Launch Vehicle. Review the status and any remaining open items or major discrepancies.

A6.1.2.6. Upper Stage (if applicable). Review the status and any remaining open items or major discrepancies.

A6.1.2.7. Satellite/Payload. Review the status and any remaining open items or major discrepancies.

A6.1.2.8. Weather Support. Review required weather support. Present DoL forecast (and back-up attempt forecast) to include standard weather parameters and probability of violation (POV), Lightening Launch Commit Criteria (LLCC) values, and as applicable; terminal area, landing site and any special weather considerations (i.e. tropical storm threats, space environmental impacts, weather equipment/communications outages, etc.).

A6.1.2.9. Range Support. Address range configuration and status of all supporting range assets and contingency plans or work-arounds. Identify "mandatory" and "required" range and safety instrumentation/support and waiver authority.

A6.1.2.10. Network Support (if applicable). Review readiness status of the Satellite Control Network (SCN), mission control centers and abort/recovery/contingency plans.

A6.1.2.11. Infrastructure. Status of critical power communications, etc.

A6.1.2.12. Safety. Present the launch hazard assessment, including an overview of the risk assessments (orbital debris, toxics, and blast overpressure), as applicable. Announce the launch mishap investigation authority. Report results of launch system safety waiver review.

A6.1.2.13. Contingency Response Support. Provide readiness status of EOC, its positioning, and key personnel. Include, by name, the ISB President, the Payload Recovery Team (PRT) Director (if applicable) and agency having mishap investigation authority. Review readiness status of the security forces and applicable waivers/deviations to security requirements.

A6.1.2.14. Countdown Summary. Review key milestones in the launch vehicle, satellite, and range countdowns for potential problem areas and hold points as well as any known collision avoidance launch countdown holds in the launch window.

A6.1.2.15. Open Action Items. Address the status of all applicable open action items (previous lessons learned, Operations Review Boards, etc.) related to the affected operation.

A6.1.2.16. Readiness. The MD, SLD/CD-O, SLD/CD-S, SLD/SE, along with FAA-AST and LSP for FAA-AST licensed launches, certify readiness to proceed.

A6.2. Day of Launch (DoL) Positions.

A6.2.1. DoL position requirements depend on many factors. Careful coordination between the spaceport, LSP or user, payload organization, and other government agencies will establish common expectations. When launching from and/or returning to DAF property, DAF representatives are essential to protect national interests and ensure proper government assessment/acceptance of inherent launch risk. SLDs should endeavor to present as common presentation of DoL positions as possible based on user, provider, and launch category.

A6.2.2. SLD/CC

A6.2.2.1. The SLD/CC (or designee) is responsible for DoL risk decisions and will collect readiness for spaceport/range systems and safety before providing launch/landing approval". **(T-2)**

A6.2.2.2. The SLD/CC will identify and document mandatory DoL positions for all launches (can be accomplished in advance by launch type) to ensure the responsibilities captured below are accomplished. **(T-2)**

A6.2.2.3. The SLD/CC (or if delegated, to the SLD/CD-O or, on day-of-launch, the individual designated with Launch Decision Authority) exercises waiver authority for mandatory launch safety requirements and public safety requirements for USG and T&E activity. For FAA-AST licensed missions, provide Range "RED/GREEN" status to the

FAA and User to meet the inherent responsibility of safe operations at their federal spaceport. **(T-2)**

A6.2.2.4. The SLD/CC (or designee) ensures that people and resources are protected to an acceptable level. **(T-2)**

A6.2.2.5. For all launch categories, ensures DoL procedures clearly identify all crewmembers. All crew members will be trained in and/or rehearse DoL procedures. Responsibilities for area surveillance, instrumentation, weather, and/or safety system readiness should be indicated by position in all procedures. **(T-2)**

A6.2.2.6. All SLD land area clearance operations will be performed by government or SLD contractor personnel. **(T-2)**

A6.2.3. Chief of Safety (SE) Responsibilities:

A6.2.3.1. Assesses safety risks and provides safety recommendations consistent with SLD/CC designated responsibilities.

A6.2.3.2. Verifies vehicle flight safety and ground support systems are ready to support launch safety functions for all non-FAA-AST licensed missions. For FAA-AST licensed missions launching from or returning to their spaceports, ranges, or installations provide status of user-requested systems/capabilities IAW applicable support agreements.

A6.2.4. Range Operations Commander (ROC) Responsibilities:

A6.2.4.1. Responsible for launch-specific mission plans for spaceport/range support. **(T-2)**

A6.2.4.2. Determines spaceport/range status and reports spaceport/range readiness based on range instrumentation and flight control status. **(T-2)**

A6.2.4.3. Ensures the proper DoL frequency mitigation procedures are implemented, as required. For GPS deconfliction issues will exercise “CEASE BUZZER” consistent with SLD developed procedures. **(T-2)**

A6.2.4.4. Coordinates status and actions with the MD/launch agency. **(T-2)**

A6.2.4.5. Directs instrumentation configuration actions and controls range timing equipment. **(T-2)**

A6.2.4.6. Conducts spaceport/range anomaly resolution actions and assesses operational impacts. **(T-2)**

A6.2.4.7. Coordinates range anomaly actions between the launch agency, the range team, and contractor personnel IAW the launch countdown procedure. **(T-3)**

A6.2.4.8. Holds count for safety or launch agency requirements, when required. **(T-2)**

A6.2.4.9. Primary DoL interface providing spaceport/range status and readiness. Provides range user with final “range green” as authorized by the SLD/CC. **(T-2)**

A6.2.4.10. Provides countdown status and reports non-nominal/errant vehicle flight and associated vehicle destruct data (Time of destruct, present position, impact position, etc) to the Missile Warning Center for notification to the National Military Command Center (NMCC).

A6.2.5. Senior/Mission Flight Control Officer (S/MFCO) Responsibilities:

A6.2.5.1. S/MFCO responsibilities are mandatory for launches utilizing USSF ground-commanded FTS and not required for AFSS. **(T-2)**

A6.2.5.2. Provide final decision for in-flight actions for launches utilizing ground-commanded FTS. The SMFCO will assess MFCO in-flight termination recommendations unless certain time-critical circumstances do not permit the MFCO and SMFCO to coordinate their actions. **(T-2)**

A6.2.5.3. Communicates directly with the SLD/CC (or designee) in the plus-count. **(T-2)**

A6.2.5.4. Execute mission flight rules consistent with SLD Safety requirements and operational needs. **(T-2)**

A6.2.5.5. Provides final Flight Control “Green/Red” for launch recommendation to the ROC in compliance with mission rules and safety requirements. **(T-2)**

A6.2.5.6. Holds count when range safety constraints are violated for ground-commanded FTS launches. **(T-2)**

A6.2.6. Area Surveillance responsibilities:

A6.2.6.1. SLDs will perform area surveillance duties for USG and T&E launches and as requested for FAA-AST licensed launches. **(T-2)**

A6.2.6.2. Directs mission-related aircraft operations conducted during launch operations. **(T-2)**

A6.2.6.3. Ensures air controllers comply with applicable DoW, Air Force, and FAA rules and procedures. **(T-2)**

A6.2.6.4. Performs launch area surveillance, which encompasses air and sea areas designated as hazard areas and the Launch Danger Zones. **(T-2)**

A6.2.6.5. Ensures hazard areas are clear of unauthorized aircraft, water-borne vessels, trains, and people prior to giving a final "GO" call for launch. **(T-2)**

A6.2.7. Launch Weather Officer (LWO) Responsibilities:

A6.2.7.1. LWO responsibilities are mandatory. The SLD will perform for USG and T&E launches. For FAA-AST licensed launches, the SLD will perform unless LSP coordinates alternate service with, and is approved by SLD/CC. **(T-3)**

A6.2.7.2. In coordination with range maintenance, weather service providers and/or the Radar, Airfield, and Weather Systems (RAWS) flight, verify weather systems are available to meet launch requirements as specified in operations directives, instructions, plans, and procedures. Provide recommendations to SLD/CC (or designee) and Safety to waive mandatory NMC weather requirements and equipment, when necessary. **(T-2)**

A6.2.7.3. Serves as the single point of contact and calls a weather “hold” when weather violations exist and/or loss of mandatory weather assets. **(T-2)**

A6.2.7.4. Observe, forecast, and evaluate weather IAW with respective Operations Directives and RSORS and upgrade appropriate weather assets to mandatory as necessary. **(T-2)**

A6.2.8. Range Engineer (RE) Responsibilities:

A6.2.8.1. RE responsibilities are mandatory for USG, T&E and ground-commanded FTS launches. **(T-2)**

A6.2.8.2. Ensures range instrumentation is available and ready to satisfy SLD safety and user requirements during launch operations. **(T-2)**

A6.2.8.3. Serves as the SLD technical expert on instrumentation, assessing the impacts of instrumentation failures, overseeing restoral actions, and committing restored systems to operational use. **(T-2)**

A6.2.8.4. Authorizes emergency configuration changes to instrumentation to meet mission requirements. **(T-2)**

Attachment 7

FREQUENCY MITIGATION & GPS TESTS, TRAINING, AND EXERCISE (TT&E) DECONFLICTION PROCESS

Table A7.1. Space Warning Squadron (SWS) Mitigation Process Steps.

STEP 1	SLD 30/45 identifies any launch requiring mitigation based on their own criteria and notifies the appropriate Range Operations and Launch Squadron at least 4 weeks prior to scheduled launch date.
STEP 2	SLD 30/45 Range Operations notifies Combined Space Operations Center (CSpOC) Strategy and Plans Division of potential conflict with at least 3 weeks' notice at: Commercial: (805) 606-9961 or DSN: 275-4135/3522/3514/4139
STEP 3	CSpOC Strategy and Plans Division confirms with appropriate Space Warning Squadron that mitigation procedures are possible for the circumstances of that particular launch and seeks Combined Joint Force Space Component Commander (CJFSCC) approval for an entry into the Combined Space Tasking Order (CSTO). The CSTO directs mitigation procedures and authorizes Direct Liaison Authorized (DIRLAUTH) between the range and the SWS concerned.
STEP 4	Range Operations and SWS agree on local procedures to check for real-time verification of mitigation during the countdown and notify the Radar operator quickly when mission is complete, or mitigation is no longer required.
STEP 5	If launch scrubs: Range advises the SWS, CSpOC Combat Ops, and CSpOC Combat Plans of the new launch date/time. Process repeats beginning with step 3.
WARNING: The following information will be classified **SECRET** **Dates of approved mitigation or verifying that a request for mitigation has been granted for a particular launch.** ***Questions regarding this warning should be directed to SFPAE/SA S3/5.*** **Note:** This SSCI is directive only to SSC units. However, launches from other ranges must work through the appropriate lead range to submit mitigation requests to the Combined Space Operations Center (CSpOC) in the same manner.	

Table A7.2. GPS TT&E Deconfliction Process.

STEP 1	No later than 10 calendar days prior to a scheduled launch mission, the SLD 30/45 Spectrum Management Office (SMO) identifies space launch missions GPS protection requirement. The Spectrum Management Office (SMO), in coordination with SLD/SE, will check USSTRATCOM GPS Electronic Attack portal, appropriate Area Frequency Coordinator (AFC) schedules. Once a conflict is identified, utilize Joint Navigational Warfare Center (JNWC) and/or GPS System Tool Kit (STK) analysis tools determine which, if any, GPS TT&E events that cause possible GPS interference conflicts, encompassing pre-launch activity through Range End-of Mission, including booster flyback.
STEP 2	When a scheduled GPS TT&E conflict is identified, the SLD 30/45 SMO (or range commander's designated alternate) will coordinate with local AFC utilizing JNWC and/or GPS STK analysis results.
STEP 3	If a critical Safety of Flight or mission risk is identified, SLD SMO or AFC representative will coordinate with the conflicting TT&E organizations IAW CJCSM 3212.03B, Enclosure D. ROPS/CC and GPS TT&E events leadership coordination may be required in determining if the conflicting GPS TT&E event should take priority over the launch mission or establish minimum stand-down times to allow accomplishment of both missions.
STEP 4	If mission requirement deconfliction cannot be achieved the ROPS/CC will coordinate with SLD/CC prior to a CEASE BUZZER notification, (requesting suspension or termination of GPS TT&E activities on day of launch) to GPS TT&E event POC IAW CJCSM 3212.03B, Enclosure D, para 2.b. and 2.d.
STEP 5	Within **72 hours** of a CEASE BUZZER notification, Range Operations will submit details (including but not limited to, GPS TT&E Event identification, contacted individual's information, launch mission identification, CEASE BUZZER caller's information, reason for CEASE BUZZER, and time/duration of suspension) to HQ CFC and SSC SMO.

Attachment 8

LOCALLY DEVELOPED RSLP PROCEDURES

A8.1. Approval Process: Use the following approval process for all Rocket Systems Launch Program (RSLP) operations involving SLS resources and a similar process for all launch campaign activities not involving SLS resources:

A8.1.1. Write local procedures with inputs from subject matter experts and engineers based upon applicable Air Force Technical Orders (TO) or other technical data.

A8.1.2. Review and update procedures with the program office, safety personnel, range/base safety office(s) and with external subject matter experts (as applicable).

A8.1.3. Incorporate changes from dry-runs, real-time redlines and recommendations from coordination.

A8.1.4. Gain procedure endorsement by the SLS/CC, SYD 80/AMX engineering, SLD/SE, and launch site safety (launch site other than SSC owned spaceports) concurrence.

A8.1.5. Gain SYD 80/AM System Program Manager approval for use of procedures endorsed by SLS/CC for the appropriate launch campaign.

A8.1.6. Conduct review of reference technical data (as applicable) immediately prior to performing the procedure for relevant changes.

A8.1.7. For potentially hazardous conditions not covered by procedures, convene a quorum of the above approval authorities to develop procedures to return resources to a safe condition and resume normal operations.

Attachment 9

CRITICAL ASSET PROTECTION

A9.1. Responsibilities.

A9.1.1. SLDs will implement common critical asset protection processes, tools, and parameters. **(T-2)**

A9.1.2. SLD Safety Offices will calculate probability of impact to protect resources on the installation from the hazards of launch, landing, and reentry. Calculations will consider the susceptibility of the facility/asset at risk to the hazard(s) presented by the launch vehicle throughout flight, and national interest of a specific launch/space vehicle. First flight, new trajectory, changed launch vehicle configurations, etc. have the potential to introduce new risks and require additional consideration. In such cases calculations should be performed as early as possible to provide maximum response time from impacted asset owners. Re-flights of existing vehicle configurations and/or trajectories should present understandable risk and might not require any calculations be performed. Concerned asset owners can engage SLD/SEs for an analysis at any time if conditions used in current analysis change.

A9.2. Calculation Process.

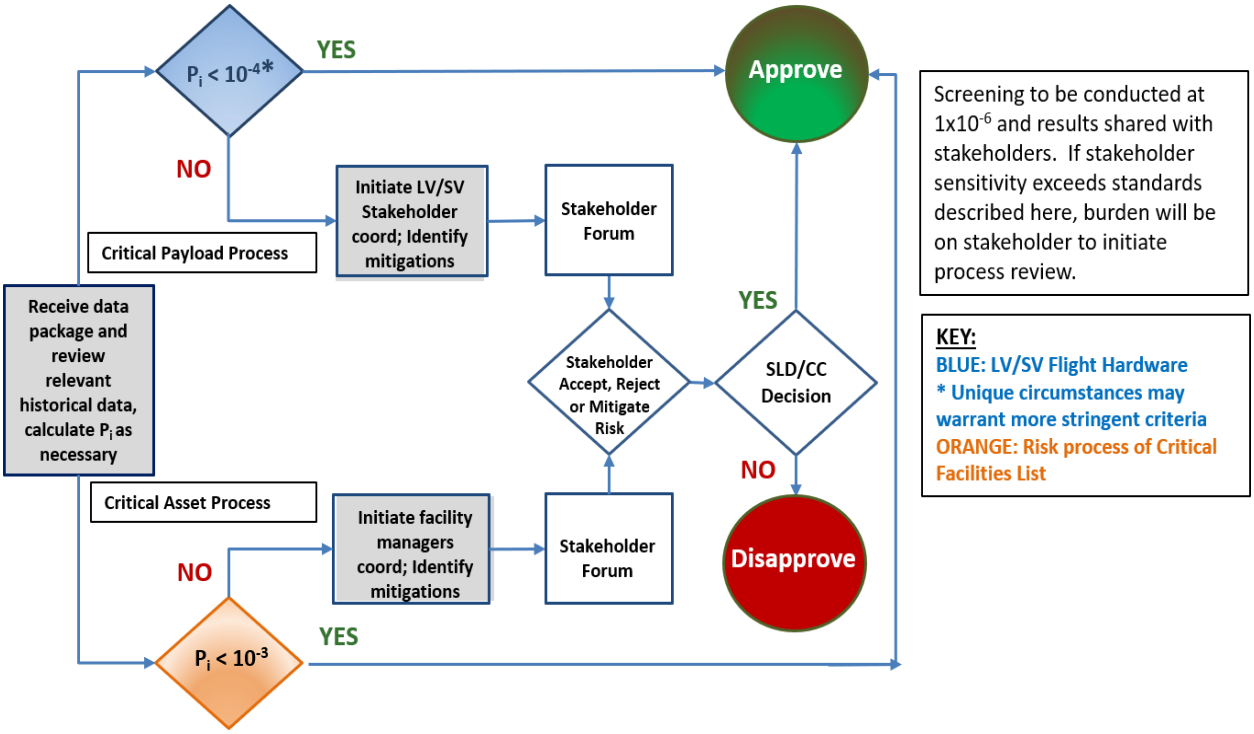
A9.2.1. Risk calculation will be performed as part of the flight safety assessment process and screen for a risk of a Probability of Impact (P_i) of $>1x10^{-6}$. This screening will provide comprehensive information to facility/asset owners to assess acceptable level of risk. The acceptable level of risk for critical assets will be a probability of $<1x10^{-3}$ (DAFI 91-202) of the risk event resulting in debris of sufficient energy penetrating the facility and the probability of damaging the facility due to nearby impacts of explosive fragments that can cause a blast overpressure producing a loss of functionality (as described in RCC 321). For critical payloads and related essential infrastructure directly supporting such a payload, a probability of $<1x10^{-4}$ of an event resulting in a debris of sufficient energy to cause the loss of the vehicle/spacecraft, while contained in a building or protected by a launch support structure, is the normal threshold of acceptable risk. The SLD Commander, in coordination with the critical payload owner, may apply down to a probability of $<1x10^{-6}$ for acceptable risk for resource protection criteria when an unsheltered critical asset, (such as but not limited to) on a launch complex, is put at risk from a third-party launch. Note: Critical asset and critical payload risk criteria above is consistent with requirements for FAA-AST licensed launches.

A9.2.2. Calculation results will be shared with launch infrastructure and asset owners as soon as information is available but NLT L-7 days, when possible. **(T-3)** As addressed above, new/changed vehicles and/or flight profiles require early calculation to allow asset owners maximum time to understand risk and possible acceptance or mitigation approaches between asset owners and launch providers. When conducting early analysis, to include at L-7 days, statistical weather data will be utilized. Forecasted weather closer to day-of-launch may result in different results. Any relevant historical data should also be included in discussions with asset owners. Probability of Impact will not be calculated on Day of Launch, and all analysis and decisions should be completed prior to the LRR. Direct engagement with stakeholders will better allow informed discussion on how actual weather will impact acceptance/mitigation decisions. Asset owners are responsible for assessing the results against the standard based on facilities/assets at risk. Asset owners will keep SLD Safety Office apprised of changes in

facility characteristics/footprint/asset impact sensitivity (i.e. amount of energy required to cause a moderate level of damage to facility/asset) or transient asset location/movement.

A9.2.3. If assets will face an unacceptable level of risk, the SLD Safety Office will hold a stakeholder forum with the activity and property owners to address specific impacts and possible mitigations. If unable to resolve the discrepancy, the stakeholder forum will present results to the SLD Commander for a decision. If a stakeholder disagrees with the Commander's decision, they will engage through their leadership channels. If significant scheduling actions or conflicts cannot be resolved at a lower level, then an Out-of-Cycle (OOC) CLSRB will convene to address concerns.

Figure A9.1. Critical Asset Protection Flow Diagram.



Attachment 10

SV COMPATIBILITY WITH NSSL

A10.1. SV Compatibility.

A10.1.1. For all NSSL missions, all USSF SVs, including integrated payload stacks of multi-manifested SV and associated adapters and interfaces, shall meet the following compatibility requirements:

A10.1.1.1. NSSL System Performance Requirements Document (SPRD) Compatibility: USSF SVs shall be compatible with the NSSL program SPRD, in particular the Mass-to-Orbit and Orbital Parameter Accuracy requirements.

A10.1.1.2. NSSL Standard Interface Specification (SIS) Compatibility: USSF SVs shall be compatible with the NSSL program SIS.

A10.1.1.3. NSSL Mission Unique Requirement Compatibility: USSF SVs with necessary mission-unique requirements shall be compatible with all NSSL LV families.

A10.1.1.4. NSSL Orbital Debris Mitigation Standard Practices (ODMSP) Compliance: The design of USSF SVs shall not preclude LV compliance with ODMSP for all NSSL LV families.

A10.1.2. For missions utilizing NSSL Phase 3 and beyond launch services (projected starting in FY 2027), it is the SSC Commander's intent that USSF SVs, including integrated payload stacks of multi-manifested SVs and associated adapters and interfaces, shall have the ability to be remanifested to another NSSL LV family as late as 12 months prior to launch. This may require USSF SVs to execute integration studies and tasks with multiple NSSL LV families.

A10.1.3. All USSF SV programs will incorporate A10.1.1. and A10.1.2. requirements into their planning and program baselines and report compliance at Program Acquisition Executive reviews. Any waivers or deviation packages require Service Acquisition Executive (SAE) concurrence prior to formal submission.

A10.1.4. To enable Space Access and operational flexibility for all National Security Space missions, all non-USSF SVs utilizing NSSL launch services should implement A10.1.1. and A10.1.2. requirements with associated effectivities.